PHYSICS

MAGNETIC EFFECT OF CURRENT_MEGA_XII



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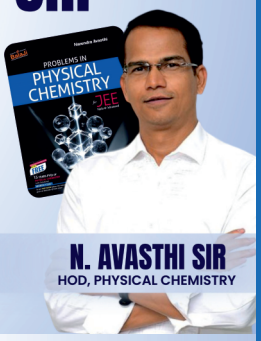

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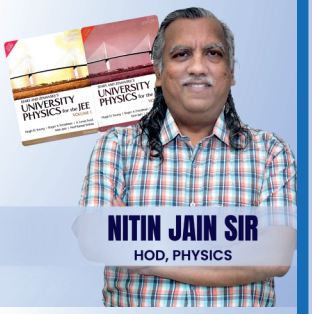


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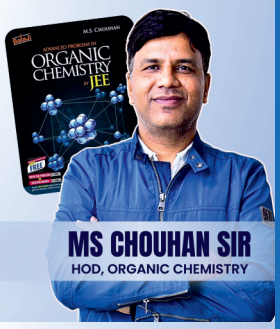


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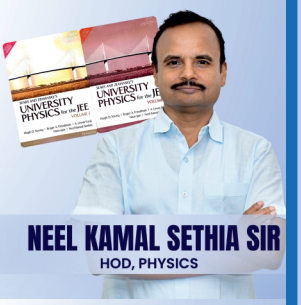


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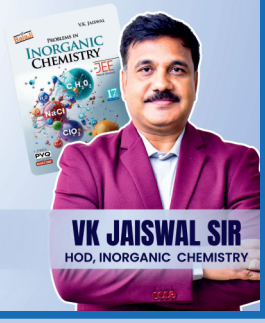


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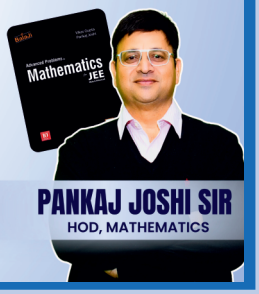


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PHYSICS

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MEGA-XII

MAGNETIC EFFECT OF CURRENT

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KEY CONCEPTS

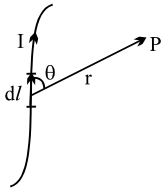
1. A static charge produces only electric field and only electric field can exert a force on it. A moving charge produces both electric field ans magnetic field and both electric field and magnetic field can exert force on it. A current carrying conductor produces only magnetic field and only magnetic field can exert a force on it.

2. Magnetic charge (i.e. current) , produces a magnetic field. It can not produce electric field as net charge on a current carrying conductor is zero. A magnetic field is detected by its action on current carrying conductors (or moving charges) and magnetic needles (compass) needles. The vector quantity $\vec{B}$ known as **MAGNETIC INDUCTION** is introduced to characterise a magnetic field. It is a vector quantity which may be defined in terms of the force it produces on electric currents. Lines of magnetic induction may be drawn in the same way as lines of electric field. The number of lines per unit area crossing a small area perpendicular to the direction of the induction bring numerically equal to $\vec{B}$. The number of lines of $\vec{B}$ crossing a given area is referred to as the **MAGNETIC FLUX** linked with that area. For this reason $\vec{B}$ is also called **MAGNETIC FLUX DENSITY** .

3 Magnetic Induction Produced By A Current (Biot-Savart Law) :

The magnetic induction dB produced by an element d*l* carrying a current I at a distance r is given by :

$$dB = \frac{\mu_0\mu_r}{4\pi}\frac{Id\ell\sin\theta}{r^2} \quad \text{or} \quad \vec{dB} = \frac{\mu_0\mu_r I(d\vec{\ell}\times\vec{r})}{4\pi\ r^3},$$



here the quantity $Id\ell$ is called as current element strength.

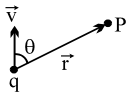
μ = permeability of the medium = $\mu_0\mu_r$; μ_0 = permeability of free space

μ_r = relative permeability of the medium (Dimensionless quantity).

Unit of μ_0 & μ is NA^{-2} or Hm^{-1} ; $\mu_0 = 4\pi \times 10^{-7}$ Hm^{-1}

4 Magnetic Induction due to a moving charge

$$dB_P = \frac{\mu_0 qv\sin\theta}{4\pi r^2}$$



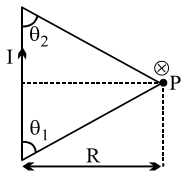
In vector form it can be written as $\vec{dB} = \frac{\mu_0}{4\pi}\frac{q(\vec{v}\times\vec{r})}{r^3}$

5. Magnetic Induction due to an inifinite st. conductor $B = \frac{\mu_0 I}{2\pi r}$

6. Magnetic Induction due to semi inifinite st. conductor $B = \frac{\mu_0 I}{4\pi r}$

7. Magnetic Induction Due To A Current Carrying Straight Conductor

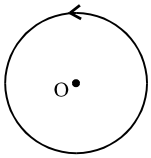
$$B = \frac{\mu_0 I}{4\pi R}(\cos\theta_1 + \cos\theta_2)$$



If the wire is very long $\theta_1 \cong \theta_2 \cong 0°$ then , $B = \frac{\mu_0 I}{2\pi R}$

8. Magnetic Field Due To A Flat Circular Coil Carrying A Current :

(i) At its centre $B = \frac{\mu_0 NI}{2R}$, direction ⊙

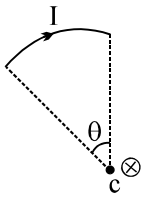


Where N = total number of turns in the coil

I = current in the coil ; R = Radius of the coil

(ii) On the axis $B = \frac{\mu_0 NIR^2}{2(x^2+R^2)^{3/2}}$

Where x = distance of the point from the centre. It is maximum at the centre.



9. Magnetic Induction due to Flat Circular Arc $B = \frac{\mu_0 I\theta}{4\pi R}$

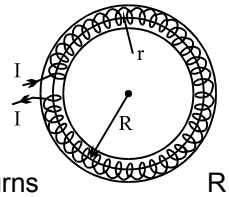
10. Magnetic Induction due to Solenoid

$B = \mu_0 nI$, direction along axis, where $n \rightarrow$ no. of turns per m. ; $I \rightarrow$ current

11. Magnetic Induction due to Toroid

$$B = \mu_0 nI$$

where $n = \frac{N}{2\pi R}$ (no. of turns per m)

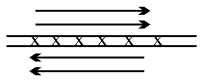


N = total turns R >> r

12. Magnetic Induction due to Current Carrying Sheet

$$B = \frac{1}{2}\mu_0 I$$

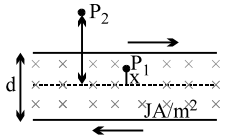
where I = Linear current density (A/m)



13. Magnetic Induction due to Thick Sheet

At point P_2 $\quad B_{out} = \frac{1}{2}\mu_0 Id$

At point P_1 $\quad B_{in} = \mu_0 Jx$



14. Gilbert's Magnetism (Earth's Magnetic Field) :

(a) The line of earth's magnetic induction lies in a vetical plane coinciding with the magnetic North - South direction at that place. This plane is called the **MAGNETIC MERIDIAN**. Earth's magnetic axis is slightly inclined to the geometric axis of earth and this angle varies from 10.5^0 to 20^0. The Earth's Magnetic poles are opposite to the geometric poles i.e. at earth's north pole, its magnetic south pole is situated and vice versa.

(b) On the magnetic meridian plane, the magnetic induction vector of the earth at any point, generally inclined to the horizontal at an angle called the **MAGNETIC DIP** at that place , such that $\vec{B}$ = total magnetic induction of the earth at that point.

$\vec{B}_v$ = the vertical component of $\vec{B}$ in the magnetic meridian plane = $B \sin\theta$.

B_H = the horizontal component of $\vec{B}$ in the magnetic meridian plane = $B \cos\theta$.

$\frac{B_v}{B_H} = \tan\theta$.

(c) At a given place on the surface of the earth , the magnetic meridian and the geographic meridian may not coincide . The angle between them is called **"DECLINATION AT THAT PLACE"** .

15 Amperes Law $\oint \vec{B} \cdot \vec{d\ell} = \mu \Sigma I$

ΣI = algebric sum of all the currents .

16. Lorentz Force :

An electric charge 'q' moving with a velocity $\vec{V}$ through a magnetic field of magnetic induction $\vec{B}$ experiences a force $\vec{F}$, given by $\vec{F} = q\vec{V} \times \vec{B}$. There fore, if the charge moves in a space where both electric and magnetic fields are superposed.

$\vec{F}$ = net electromagnetic force on the charge = $q\vec{E} + q\vec{V} \times \vec{B}$

This force is called the **LORENTZ FORCE.**

17. Motion of A charge in uniform magnetic field :

(a) When $\vec{v}$ is || to $\vec{B}$: Motion will be in a st. line and $\vec{F}$ = 0

(b) When $\vec{v}$ is $\perp$ to $\vec{B}$: Motion will be in circular path with radius $R = \frac{mv}{qB}$ and angular velocity $\omega = \frac{qB}{m}$ and $F = qvB$.

(c) When $\vec{v}$ is at $\angle\theta$ to $\vec{B}$: Motion will be helical with radius $R_k = \frac{mv\sin\theta}{qB}$ and pitch $P_H = \frac{2\pi mv\cos\theta}{qB}$ and $F = qvB\sin\theta$.

18. Magnetic Force On A Straight Current Carrying Wire :

$\vec{F} = I(\vec{L} \times \vec{B})$

I = current in the straight conductor

$\vec{L}$ = length of the conductor in the direction of the current in it

$\vec{B}$ = magnetic induction. (Uniform throughout the length of conduction)

Note : In general force is $\vec{F} = \int I(d\vec{\ell} \times \vec{B})$

19. Magnetic Interaction Force Between Two Parallel Long Straight Currents :

When two long straight linear conductors are parallel and carry a current in each, they magnetically interact with each other, one experiences a force. This force is of :

(i) Repulsion if the currents are anti-parallel (i.e. in opposite direction) or

(ii) Attraction if the currents are parallel (i.e. in the same direction)

This force per unit length on either conductor is given by $F = \frac{\mu_0}{2\pi}\frac{I_1 I_2}{r}$. Where r = perpendicular distance between the parallel conductors

20. Magnetic Torque On A Closed Current Circuit :

When a plane closed current circuit of 'N' turns and of area 'A' per turn carrying a current I is placed in uniform magnetic field, it experience a zero nett force, but experience a torque given by $\vec{\tau} = NI\vec{A} \times \vec{B} = \vec{M} \times \vec{B} = BINA\sin\theta$. When $\vec{A}$ = area vector outward from the face of the circuit where the current is anticlockwise, $\vec{B}$ = magnetic induction of the uniform magnetic feild. $\vec{M}$ = magnetic moment of the current circuit = $IN\vec{A}$

Note : This expression can be used only if $\vec{B}$ is uniform otherwise calculus will be used.

21. Moving Coil Galvanometer :

It consists of a plane coil of many turns suspended in a radial magnetic feild. when a current is passed in the coil it experiences a torque which produces a twist in the suspension. This deflection is directly proportional to the torque ∴ NIAB = Kθ

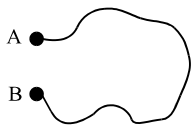
$I = \left(\frac{K}{NAB}\right)\theta$ K = elastic torsional constant of the suspension

I = C θ $C C = \frac{K}{NAB}$ = GALVANOMETER CONSTANT.

22. Force Experienced By A Magnetic Dipole In A Non-Uniform Magnetic Field :

$|\vec{F}| = \left|M\frac{\partial B}{\partial r}\right|$ where M = Magnetic dipole moment.

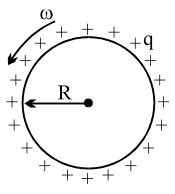
23. Force on a random shaped conductor in magnetic field :



1. Magnetic force on a loop in a uniform $\vec{B}$ is zero

2. Force experienced by a wire of any shape is equivalent to force on a wire joining points A & B in a uniform magnetic field .

24. Magnetic moment of a rotating charge :



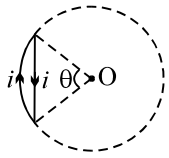
If a charge q is rotating at an angular velocity ω, its equivalent current is given as $I = \frac{q\omega}{2\pi}$ & its magnetic moment is $M = I\pi R^2 = 1/2 q\omega R^2$.

NOTE : The rate of magnetic moment to Angular momentum of a uniform rotating object which is charged uniformly is always a constant. Irrespective of the shape of conductor M/L = q/2m

EXERCISE-I

ONLY ONE OPTION IS CORRECT

1. Net magnetic field at the centre of the circle O due to a current carrying loop as shown in figure is ($\theta < 180°$)



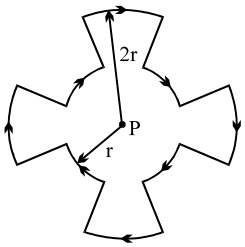
(A) zero

(B) perpendicular to paper inwards

(C) perpendicular to paper outwards

(D) is perpendicular to paper inwards if $\theta \le 90°$ and perpendicular to paper outwards if $90° \le \theta < 180°$

2. A current I flows around a closed path in the horizontal plane of the circle as shown in the figure. The path consists of eight arcs with alternating radii r and 2r. Each segment of arc subtends equal angle at the common centre P. The magnetic field produced by current path at point P is



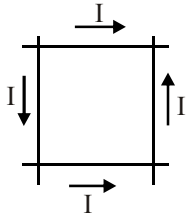
(A) $\frac{3}{8}\frac{\mu_0 I}{r}$; perpendicular to the plane of the paper and directed inward.

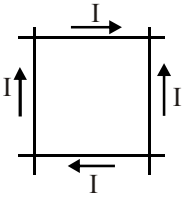
(B) $\frac{3}{8}\frac{\mu_0 I}{r}$; perpendicular to the plane of the paper and directed outward.

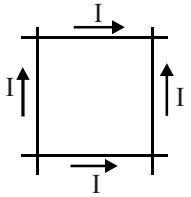
(C) $\frac{1}{8}\frac{\mu_0 I}{r}$; perpendicular to the plane of the paper and directed inward.

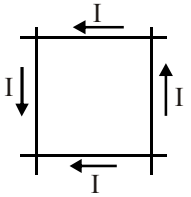
(D) $\frac{1}{8}\frac{\mu_0 I}{r}$; perpendicular to the plane of the paper and directed outward.

3. The figure shows four different sets of wires that cross each other without actually touching. The magnitude of the current is the same in all four cases, and the directions of current flow are as indicated. For which configuration will the magnetic field at the center of the square formed by the wires be equal to zero?

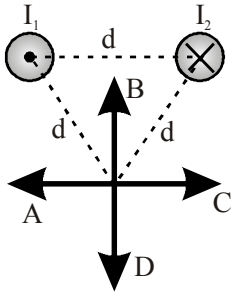
(A) 

(B) 

(C) 

(D) 

4. The figure shows two long wires carrying equal currents I_1 and I_2 flowing in opposite directions. Which of the arrows labeled A to D correctly represents the direction of the magnetic field due to the wires at a point located at an equal distance d from each wire?

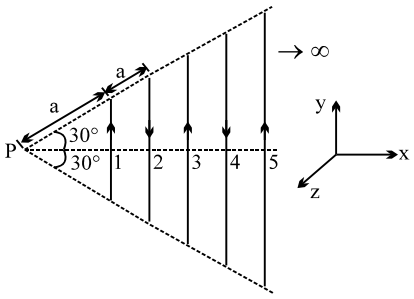


(A) A (B) B (C) C (D) D

5. A direct current is passing through a wire. It is bent to form a coil of one turn. Now it is further bent to form a coil of two turns but at smaller radius. The ratio of the magnetic induction at the centre of this coil and at the centre of the coil of one turn is

(A) 1 : 4 (B) 4 : 1 (C) 2 : 1 (D) 1 : 1

6. Infinite number of straight wires each carrying current I are equally placed as shown in the figure. Adjacent wires have current in opposite direction. Net magnetic field at point P is



(A) $\frac{\mu_0 I}{4\pi}\frac{l\text{n}\,2}{\sqrt{3}\,a}\hat{k}$ (B) $\frac{\mu_0 I}{4\pi}\frac{l\text{n}\,4}{\sqrt{3}\,a}\hat{k}$ (C) $\frac{\mu_0 I}{4\pi}\frac{l\text{n}\,4}{\sqrt{3}\,a}(-\hat{k})$ (D) Zero

7. Two mutually perpendicular conductors carrying currents I_1 and I_2 lie in one plane. Locus of the point at which the magnetic induction is zero, is a

(A) circle with centre as the point of intersection of the conductor.

(B) parabola with vertex as the point of intersection of the conductors

(C) straight line passing through the point of intersection of the conductors.

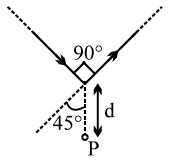
(D) rectangular hyperbola

8. Two concentric coils X and Y of radii 16 cm and 10 cm lie in the same vertical plane containing N-S direction. X has 20 turns and carries 16 A. Y has 25 turns & carries 18A. X has current in anticlockwise direction and Y has current in clockwise direction for an observer, looking at the coils facing the west. The magnitude of net magnetic field at their common centre is

(A) $5\pi \times 10^{-4}$ T towards west (B) $13\pi \times 10^{-4}$ T towards east

(C) $13\pi \times 10^{-4}$ T towards west (D) $5\pi \times 10^{-4}$ T towards east

9. Find the magnetic field at P due to the arrangement shown

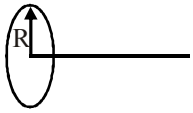


(A) $\frac{\mu_0 i}{\sqrt{2}\,\pi d}\left(1-\frac{1}{\sqrt{2}}\right)\otimes$ (B) $\frac{2\mu_0 i}{\sqrt{2}\,\pi d}\otimes$ (C) $\frac{\mu_0 i}{\sqrt{2}\,\pi d}\otimes$ (D) $\frac{\mu_0 i}{\sqrt{2}\,\pi d}\left(1+\frac{1}{\sqrt{2}}\right)\otimes$

10. Equal current *i* is flowing in three infinitely long wires along positive x, y and z directions. The magnetic field at a point (0, 0, –a) would be :

(A) $\frac{\mu_0 i}{2\pi a}(\hat{j}-\hat{i})$ (B) $\frac{\mu_0 i}{2\pi a}(\hat{i}+\hat{j})$ (C) $\frac{\mu_0 i}{2\pi a}(\hat{i}-\hat{j})$ (D) $\frac{\mu_0 i}{2\pi a}(\hat{i}+\hat{j}+\hat{k})$

11. Constant current I is flowing through a circular coil of radius R. At what distance from the centre will the magnetic field (on the axis) be maximum :



(A) $\frac{R}{\sqrt{2}}$ (B) $\frac{R}{2}$

(C) R (D) zero

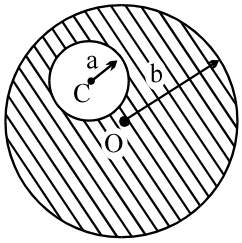
12. A hollow cylinder having infinite length and carrying uniform current per unit length λ along the circumference as shown. Magnetic field inside the cylinder is



(A) $\frac{\mu_0\lambda}{2}$ (B) $\mu_0\lambda$

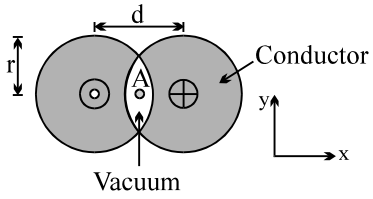
(C) $2\mu_0\lambda$ (D) none

13. A long straight metal rod has a very long hole of radius 'a' drilled parallel to the rod axis as shown in the figure. If the rod carries a current 'i' find the value of magnetic induction on the axis of the hole, where OC = c



(A) $\frac{\mu_0 ic}{\pi(b^2-a^2)}$ (B) $\frac{\mu_0 ic}{2\pi(b^2-a^2)}$ (C) $\frac{\mu_0 i(b^2-a^2)}{2\pi c}$ (D) $\frac{\mu_0 ic}{2\pi a^2 b^2}$

14. Two long conductors are arranged as shown above to form overlapping cylinders, each of raidus r, whose centers are separated by a distance d. Current of density J flows into the plane of the page along the shaded part of one conductor and an equal current flows out of the plane of the page along the shaded portion of the other, as shown. What are the magnitude and direction of the magnetic field at point A?



(A) $(\mu_0/2\pi)\pi dJ$, in the +y-direction (B) $(\mu_0/2\pi)d^2/r$, in the +y-direction

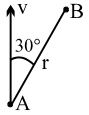
(C) $(\mu_0/2\pi)4d^2J/r$, in the –y-direction (D) $(\mu_0/2\pi)Jr^2/d$, in the –y-direction

(E) There is no magnetic field at A.

15. The dimensional formula for the physical quantity $\frac{E^2\mu_0\varepsilon_0}{B^2}$ is (E = electric field and B = magnetic field)

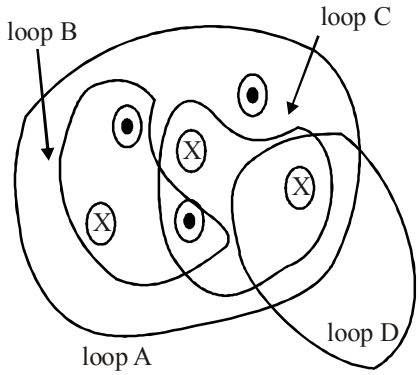
(A) $L^0M^0T^0$ (B) $L^1M^0T^{-1}$ (C) $L^{-1}M^0T^1$ (D) $L^{1/2}M^0T^{-1/2}$

16. A charge particle A of charge q = 2 C has velocity v = 100 m/s. When it passes through point A and has velocity in the direction shown. The strength of magnetic field at point B due to this moving charge is (r = 2 m).



(A) 2.5 μT (B) 5.0 μT (C) 2.0 μT (D) None

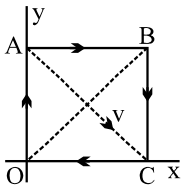
17. Consider six wires coming into or out of the page, all with the same current. Rank the line integral of the magnetic field (from most positive to most negative) taken counterclockwise around each loop shown.



(A) B > C > D > A (B) B > C = D > A (C) B > A > C = D (D) C > B = D > A

18. An electron is moving along positive x-axis. A uniform electric field exists towards negative y-axis. What should be the direction of magnetic field of suitable magnitude so that net force of electron is zero
(A) positive z- axis (B) negative z-axis (C) positive y-axis (D) negative y-axis

19. OABC is a current carrying square loop an electron is projected from the centre of loop along its diagonal AC as shown. Unit vector in the direction of initial acceleration will be



(A) $\hat{k}$ (B) $-\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

(C) $-\hat{k}$ (D) $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

20. An electron moving with a velocity $\vec{V}_1 = 2\hat{i}$ m/s at a point in a magnetic field experiences a force $\vec{F}_1 = -2\hat{j}N$.

If the electron is moving with a velocity $\vec{V}_2 = 2\hat{j}$ m/s at the same point, it experiences a force $\vec{F}_2 = +2\hat{i}N$.

The force the electron would experience if it were moving with a velocity $\vec{V}_3 = 2\hat{k}$ m/s at the same point is

(A) zero (B) $2\hat{k}N$ (C) $-2\hat{k}N$ (D) information is insufficient

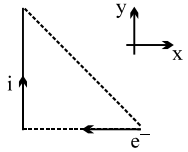
21. A conductor of length *l* and mass m is placed along the east-west line on a table. Suddenly a certain amount of charge is passed through it and it is found to jump to a height h. The earth's magnetic induction is B. The charge passed through the conductor is: (B is horizontal)

(A) $\frac{1}{Bmgh}$ (B) $\frac{\sqrt{2gh}}{B\ell m}$ (C) $\frac{gh}{B\ell m}$ (D) $\frac{m\sqrt{2gh}}{B\ell}$

22. A electron experiences a force $\left(4.0\hat{i}+3.0\hat{j}\right)$ × 10^{-13} N in a uniform magnetic field when its velocity is $2.5\hat{k}\times10^{7}$ ms^{-1}. When the velocity is redirected and becomes $\left(1.5\hat{i}-2.0\hat{j}\right)\times10^{7}$ ms^{-1}, the magnetic force of the electron is zero. The magnetic field vector $\vec{B}$ is :

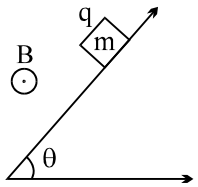
(A) $-0.075\hat{i}+0.1\hat{j}$ (B) $0.1\hat{i}+0.075\hat{j}$ (C) $0.075\hat{i}-0.1\hat{j}+\hat{k}$ (D) $0.075\hat{i}-0.1\hat{j}$

23. The direction of magnetic force on the electron as shown in the diagram is along



(A) y-axis (B) –y-axis (C) z-axis (D) –z-axis

24. A block of mass m & charge q is released on a long smooth inclined plane magnetic field B is constant, uniform, horizontal and parallel to surface as shown. Find the time from start when block loses contact with the surface.



(A) $\frac{m\cos\theta}{qB}$ (B) $\frac{m\,\mathrm{cosec}\,\theta}{qB}$

(C) $\frac{m\cot\theta}{qB}$ (D) none

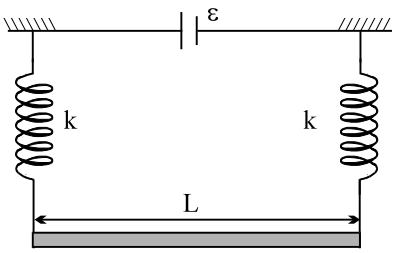
25. A straight rod of mass m and length L is suspended from the identical spring as shown in the figure. The spring stretched by a distance of x_0 due to the weight of the wire. The circuit has total resistance RΩ. When the magnetic field perpendicular to the plane of the paper is switched on, springs are observed to extend further by the same distance. The magnetic field strength is

(A) $\frac{mgR}{\varepsilon L}$; directed outward from the plane of the paper

(B) $\frac{mgR}{2\varepsilon x_0}$; directed outward from the plane of the paper

(C) $\frac{mgR}{\varepsilon L}$; directed into the plane of the paper

(D) $\frac{mgR}{\varepsilon x_0}$; directed into the plane of the paper

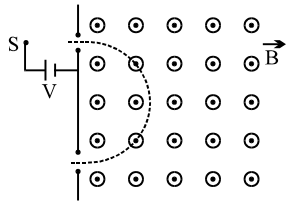


26. A charged particle is moving in the presence of electric ($\vec{E}$)and magnetic $(\vec{B})$ fields. The directions of $\vec{E}$ and $\vec{B}$ are such that the charged particle moves in a straight line and its speed increases. The relations amongst $\vec{E}$, $\vec{B}$ and velocity $\vec{V}$ must be such that

(A) $\vec{E}\,.\,\vec{B}$ = 0, $\vec{V}$ is arbitrary (B) $\vec{E}$, $\vec{B}$ and $\vec{V}$ are all parallel to each other

(C) $\vec{E}\,.\,\vec{V}$ = 0, $\vec{B}\,.\,\vec{V}$ = 0 but $\vec{E}\,.\,\vec{B}\neq0$ (D) $\vec{V}$ is parallel to $\vec{E}$ and perpendicular to $\vec{B}$

27. A mass spectrometer is a device which select particle of equal mass. An iron with electric charge q > 0 starts at rest from a source S and is accelerated through a potential difference V. It passes through a hole into a region of constant magnetic field $\vec{B}$ perpendicular to the plane of the paper as shown in the figure. The particle is deflected by the magnetic field and emerges through the bottom hole at a distance d from the top hole. The mass of the particle is



(A) $\frac{qBd}{V}$ (B) $\frac{qB^2d^2}{4V}$ (C) $\frac{qB^2d^2}{8V}$ (D) $\frac{qBd}{2V}$

28. Electrons moving with different speeds enter a uniform magnetic field in a direction perpendicular to the field. They will move along circular paths.
(A) of same radius
(B) with larger radii for the faster electrons
(C) with smaller radii for the faster electrons
(D) either (B) or (C) depending on the magnitude of the magnetic field

29. Three ions H^+, He^+ and O^{+2} having same kinetic energy pass through a region in which there width is a uniform magnetic field perpendicular to their velocity, then :
(A) H^+ will be least deflected. (B) He^+ and O^{+2} will be deflected equally.
(C) O^{+2} will be deflected most. (D) all will be deflected equally.

30. An electron having kinetic energy T is moving in a circular orbit of radius R perpendicular to a uniform magnetic induction $\vec{B}$. If kinetic energy is doubled and magnetic induction tripled, the radius will become

(A) $\frac{3R}{2}$ (B) $\sqrt{\frac{3}{2}}$ R (C) $\sqrt{\frac{2}{9}}$ R (D) $\sqrt{\frac{4}{3}}$ R

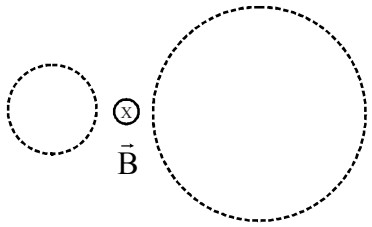
31. An electron (mass = 9.1×10^{-31} ; charge = -1.6×10^{-19} C) experiences no deflection if subjected to an electric field of 3.2×10^5 V/m and a magnetic field of 2.0×10^{-3} Wb/m^2 . Both the fields are normal to the path of electron and to each other . If the electric field is removed, then the electron will revolve in an orbit of radius
(A) 45 m (B) 4.5 m (C) 0.45 m (D) 0.045 m

32. A charged particle moves in a magnetic field $\vec{B} = 10\hat{i}$ with initial velocity $\vec{u} = 5\hat{i} + 4\hat{j}$. The path of the particle will be
(A) straight line (B) circle (C) helical (D) none

33. An electron is projected with velocity v_0 in a uniform electric field E perpendicular to the field. Again it is projetced with velocity v_0 perpendicular to a uniform magnetic field B. If r_1 is initial radius of curvature just after entering in the electric field and r_2 is initial radius of curvature just after entering in magnetic field then the ratio r_1/r_2 is equal to

(A) $\frac{Bv_0^2}{E}$ (B) $\frac{B}{E}$ (C) $\frac{Ev_0}{B}$ (D) $\frac{Bv_0}{E}$

34. An electron and a proton each travel with equal speeds around circular orbits in the same uniform magnetic field as indicated (not to scale) in the diagram. The field is into the page on the diagram. The electron travels _____ around the _____ circle and the proton travels _____ around the _____ circle.

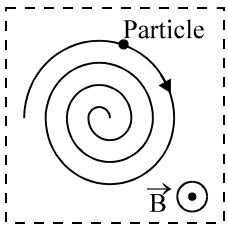


(A) clockwise, smaller, counter-clockwise, larger (B) counterclockwise, smaller, clockwise, larger
(C) clockwise, larger, counterclockwise, smaller (D) counterclockwise, larger, clockwise, smaller

35. Two particles A and B having equal charges after being accelerated through the same potential difference, enter a region of uniform magnetic field and describe circular paths of radii R_1 and R_2 respectively. The ratio of mass of A to that of B is :

(A) $(R_1 / R_2)^2$ (B) $(R_1 / R_2)^{1/2}$ (C) $(R_2 / R_1)^{3/2}$ (D) (R_1 / R_2)

36. A uniform magnetic field is directed out of the page. A charged particle, moving in the plane of the page, follows a clockwise spiral of decreasing radius as shown. A reasonable explanation is :



(A) the charge is positive and slowing down (B) the charge is negative and slowing down
(C) the charge is positive and speeding up (D) the charge is negative and speeding up

37. A charged particle is released from rest in a region of uniform electric and magnetic fields, which are parallel to each other. The locus of the particle will be

(A) helix of constant pitch (B) straight line
(C) helix of varying pitch (D) cycloid

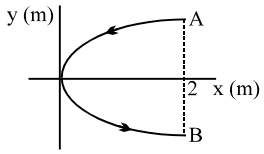
38. A particle having charge of 1 C, mass 1 kg and speed 1 m/s enters a uniform magnetic field, having magnetic induction of 1 T, at an angle $\theta = 30°$ between velocity vector and magnetic induction. The pitch of its helical path is (in meters)

(A) $\frac{\sqrt{3}\pi}{2}$ (B) $\sqrt{3}\pi$ (C) $\frac{\pi}{2}$ (D) π

39. A particle of specific charge (q/m) is projected from the origin of coordinates with initial velocity [u**i** – v**j**]. Uniform electric magnetic fields exist in the region along the +y direction, of magnitude E and B. The particle will definitely return to the origin once if

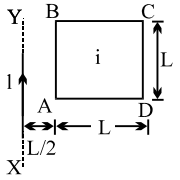
(A) $[vB/2\pi E]$ is an integer (B) $(u^2 + v^2)^{1/2}$ $[B/\pi E]$ is an integer
(C) $[vB/\pi E]$ in an integer (D) $[uB/\pi E]$ is an integer

40. A conducting wire bent in the form of a parabola $y^2 = 2x$ carries a current i = 2 A as shown in figure. This wire is placed in a uniform magnetic field $\vec{B} = -4\hat{k}$ Tesla. The magnetic force on the wire is (in newton)



(A) $-16\hat{i}$ (B) $32\hat{i}$ (C) $-32\hat{i}$ (D) $16\hat{i}$

41. A square loop ABCD, carrying a current i, is placed near and coplanar with a long straight conductor XY carrying a current I, the net force on the loop will be



(A) $\frac{2\mu_0 Ii}{3\pi}$ (B) $\frac{\mu_0 Ii}{2\pi}$

(C) $\frac{2\mu_0 Iil}{3\pi}$ (D) $\frac{\mu_0 Iil}{2\pi}$

42. A straight current carrying conductor is placed in such a way that the current in the conductor flows in the direction out of the plane of the paper. The conductor is placed between two poles of two magnets, as shown. The conductor will experience a force in the direction towards

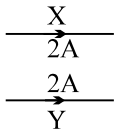


(A) P (B) Q (C) R (D) S

43. A metal ring of radius r = 0.5 m with its plane normal to a uniform magnetic field B of induction 0.2 T carries a current I = 100 A. The tension in newtons developed in the ring is :

(A) 100 (B) 50 (C) 25 (D) 10

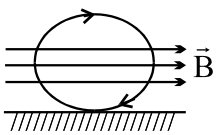
44. In given figure, X and Y are two long straight parallel conductors each carrying a current of 2 A. The force on each conductor is F newtons. When the current in each is changed to 1 A and reversed in direction, the force on each is now



(A) F/4 and unchanged in direction (B) F/2 and reversed in direction

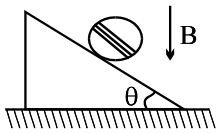
(C) F/2 and unchanged in direction (D) F/4 and reversed in direction

45. A conducting ring of mass 2 kg and radius 0.5 m is placed on a smooth horizontal plane. The ring carries a current i = 4A. A horizontal magnetic field B = 10T is switched on at time t = 0 as shown in figure. The initial angular acceleration of the ring will be



(A) 40 π rad/s^2 (B) 20 π rad/s^2 (C) 5 π rad/s^2 (D) 15 π rad/s^2

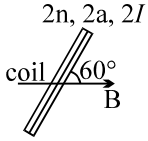
46. In the figure shown a coil of single turn is wound on a sphere of radius R and mass m. The plane of the coil is parallel to the plane and lies in the equatorial plane of the sphere. Current in the coil is i. The value of B if the sphere is in equilibrium is



(A) $\frac{mg\cos\theta}{\pi iR}$ (B) $\frac{mg}{\pi iR}$

(C) $\frac{mg\tan\theta}{\pi iR}$ (D) $\frac{mg\sin\theta}{\pi iR}$

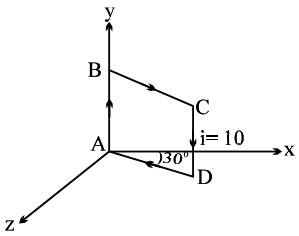
47. A rectangular coil PQ has 2n turns, an area 2a and carries a current 2*I*, (refer figure). The plane of the coil is at 60° to a horizontal uniform magnetic field of flux density B. The torque on the coil due to magnetic force is



(A) Bna*I* sin60° (B) 8Bna*I* cos60°

(C) 4na*I* Bsin60° (D) none

48. Figure shows a square current carrying loop ABCD of side 10 cm and current i = 10A. The magnetic moment $\vec{M}$ of the loop is



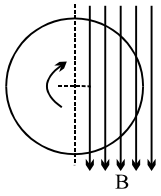
(A) (0.05) $\left(\hat{i}-\sqrt{3}\hat{k}\right)A-m^2$

(B) (0.05) $\left(\hat{j}+\hat{k}\right)A-m^2$

(C) (0.05) $\left(\sqrt{3}\hat{i}+\hat{k}\right)A-m^2$

(D) $\left(\hat{i}+\hat{k}\right)A-m^2$

49. A thin non conducting disc of radius R is rotating clockwise (see figure) with an angular velocity w about its central axis, which is perpendicular to its plane. Both its surfaces carry +ve charges of uniform surface density. Half the disc is in a region of a uniform, unidirectional magnetic field B parallel to the plane of the disc, as shown. Then,



(A) The net torque on the disc is zero

(B) The net torque vector on the disc is directed leftwards.

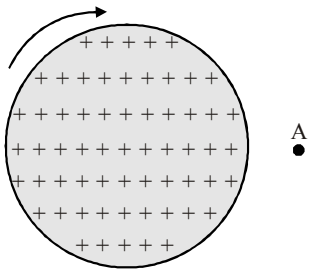
(C) The net torque vector on the disc is directed rightwards.

(D) The net torque vector on the disc is parallel to B.

50. The magnetic moment of a circular orbit of radius 'r' carrying a charge 'q' and rotating with velocity v is given by

(A) $\frac{qvr}{2\pi}$ (B) $\frac{qvr}{2}$ (C) $qv\pi r$ (D) $qv\pi r^2$

51. A positively charged disk is rotated clockwise as shown in the figure below. What is the direction of the magnetic field at point A in the plane of the disk?



(A) ⊗ (into the page)

(B) ⟶

(C) ⟵

(D) ⊙ (out of the page)

Paragraph for question nos. 52 to 54

A proton, an α particle, a doubly isonised Lithium atom $(^6Li)^{2+}$ and triply ionised carbon atom $(^{12}C)^{3+}$ are projected in a region having uniform magnetic field. Subsequently they exhibit helical motion (or circular motion depending on direction of projection) with radius r_1, r_2, r_3, r_4 and pitch P_1, P_2, P_3, P_4 respectively.

52. If each is projected perpendicular to the magnetic field with same kinetic energy, then :
(A) $r_1 = r_2 = r_3 = r_4$ (B) $r_4 > r_3 > r_2 = r_1$ (C) $r_1 > r_2 > r_3 > r_4$ (D) $r_3 > r_4 > r_2 = r_1$

53. If each is projected with same velocity each at an angle of 45° with respect to the magnetic field, then :
(A) $P_1 = P_2 = P_3 = P_4$ (B) $P_1 > P_2 > P_3 > P_4$ (C) $P_4 > P_3 > P_2 = P_1$ (D) $P_4 > P_3 > P_2 > P_1$

54. If each particle were accelerated through the same potential difference and then projected at 45° to the magnetic field then :
(A) $r_4 = r_3 = r_2 = r_1$ (B) $r_1 > r_2 > r_3 > r_4$ (C) $r_4 > r_3 > r_2 > r_1$ (D) $r_4 > r_3 > r_2 = r_1$

55. **Statement-1 :** It is not possible for a charged particle to move in a circular path around a long straight uncharged conductor carrying current under the influence of its magnetic field alone.
Statement-2 : The magnetic force (if nonzero) on a moving charged particle is normal to its velocity.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

56. **Statement-1 :** For a charged particle to pass through a uniform electro-magnetic field without change in velocity, its velocity vector must be perpendicular to the magnetic field.

Statement-2 : Net Lorentz force on the particle is given by $\vec{F} = q[\vec{E} + \vec{v} \times \vec{B}]$

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

57. **Statement-1 :** Ampere law can be used to find magnetic field due to finite length of a straight current carrying wire.
Statement-2 : The magnetic field due to finite length of a straight current carrying wire is symmetric about the wire.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

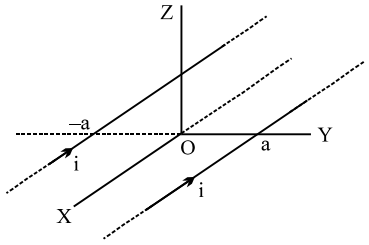
58. Consider the magnetic field produced by a finitely long current carrying wire.
(A) the lines of field will be concentric circles with centres on the wire.
(B) There can be two points in the same plane where magnetic fields are same.
(C) There can be large number of points where the magnetic field is same.
(D) The magnetic field at a point is inversally proportional to the distance of the point from the wire.

59. Consider three quantities x = E/B, y = $\sqrt{1/\mu_0\varepsilon_0}$ and z = $\frac{l}{CR}$. Here, *l* is the length of a wire, C is a capacitance and R is a resistance. All other symbols have standard meanings.

(A) x, y have the same dimensions (B) y, z have the same dimensions

(C) z, x have the same dimensions (D) none of the three pairs have the same dimensions.

60. Two long thin, parallel conductors carrying equal currents in the same direction are fixed parallel to the x-axis, one passing through y = a and the other through y = –a. The resultant magnetic field due to the two conductors at any point is B. Which of the following are correct?



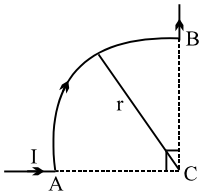
(A) B = 0 for all points on the x-axis

(B) At all points on the y-axis, excluding the origin, B has only a z-component.

(C) At all points on the z-axis, excluding the origin, B has only a y-component.

(D) B cannot have an x-component.

61. A wire carrying I is shaped as shown. Section AB is a quarter circle of radius r. The magnetic field at C is directed



(A) along the bisector of the angle ACB, away from AB

(B) along the bisector of the angle ACB, towards AB

(C) perpendicular to the plane of the paper, directed into the paper

(D) at an angle $\pi/4$ to the plane of the paper

62. A long straight wire carries a current along the x-axis. Consider the points A(0, 1, 0), B(0, 1, 1), C(1, 0, 1) and D(1, 1, 1). Which of the following pairs of points will have magnetic fields of the same magnitude?

(A) A and B (B) A and C (C) B and C (D) B and D

63. In the previous question, if the current is *i* and the magnetic field at D has magnitude B,

(A) B = $\frac{\mu_0 i}{2\sqrt{2}\pi}$ (B) B = $\frac{\mu_0 i}{2\sqrt{3}\pi}$

(C) B is parallel to the x-axis (D) B makes an angle of 45° with the xy plane

64. Which of the following statement is correct :

(A) A charged particle enters a region of uniform magnetic field at an angle 85^0 to magnetic lines of force. The path of the particle is a circle.

(B) An electron and proton are moving with the same kinetic energy along the same direction. When they pass through uniform magnetic field perpendicular to their direction of motion, they describe circular path.

(C) There is no change in the energy of a charged particle moving in a magnetic field although magnetic force acts on it.

(D) Two electrons enter with the same speed but in opposite direction in a uniform transverse magnetic field. Then the two describe circle of the same radius and these move in the same direction.

65. Consider the following statements regarding a charged particle in a magnetic field. Which of the statements are true :

(A) Starting with zero velocity, it accelerates in a direction perpendicular to the magnetic field.

(B) While deflecting in magnetic field its energy gradually increases .

(C) Only the component of magnetic field perpendicular to the direction of motion of the charged particle is effective in deflecting it.

(D) Direction of deflecting force on the moving charged particle is perpendicular to its velocity.

66. A particle of charge q and velocity v passes with constant velocity through a space with non-zero electric field E and magnetic field B. The undeflecting conditions will hold if.

(A) signs of both q and E are reversed.

(B) signs of both q and B are reversed.

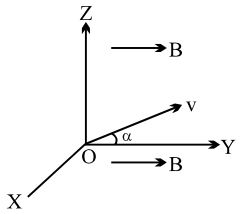
(C) both B and E are changed in magnitude, but keeping the product of |B| and |E| fixed.

(D) both B and E are doubled in magnitude.

67. An electron is moving along the positive X-axis. You want to apply a magnetic field for a short time so that the electron may reverse its direction and move parallel to the negative X–axis. This can be done by applying the magnetic field along

(A) Y-axis (B) Z-axis (C) Y-axis only (D) Z-axis only

68. In a region of space, a uniform magnetic field B exists in the y-direction. A proton is fired from the origin, with its initial velocity v making a small angle α with the y-direction in the yz plane. In the subsequent motion of the proton,



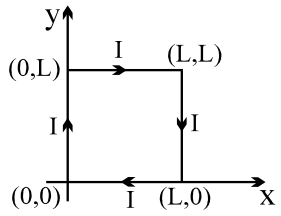
(A) its x-coordinate can never be positive

(B) its x- and z-coordinates cannot both be zero at the same time

(C) its z-coordinate can never be negative

(D) its y-coordinate will be proportional to the square of its time of flight

69. Figure shows a square loop in x, y plane carrying current I present in the magnetic field which is given by $\vec{B} = \frac{B_0 z}{L}\hat{j} + \frac{B_0 y}{L}\hat{k}$ where B_0 is positive constant. Which of the following statement(s) is/are **correct**?



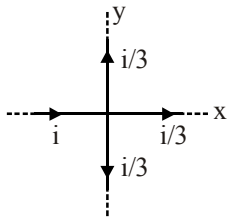
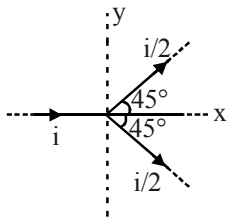
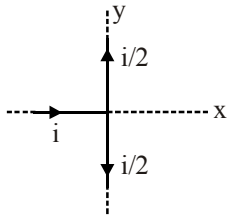
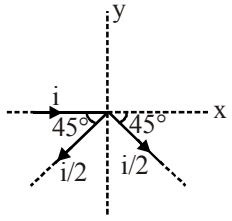
(A) force on side (0,0) to (0, L) is $\left(\frac{B_0 I L}{2}\right)\hat{i}$. (B) force on side (0, L) to (L, L) is $-B_0 I L \hat{j}$.

(C) Net magnetic force on loop is zero. (D) force on side (L, 0) to (0, 0) is zero.

70. A charged particle having non zero velocity is subjected to certain conditions given in column-I, column-II gives possible trajectories of the particle. Match the conditions in column-I with the results in column-II.

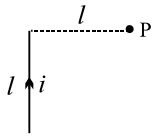
	Column-I		**Column-II**
(A)	In only uniform electric field	(P)	the path of the charged particle may be a straight line
(B)	In only uniform magnetic field	(Q)	the path of the charged particle may be a parabola
(C)	In uniform magnetic and uniform electric field	(R)	the path of the charged particle may be circular
(D)	Subjected to a net force of constant magnitude	(S)	the path of the charged particle may be a helical with uniform or non uniform pitch

71. Column-II gives four situations in which three or four semi infinite current carrying wires are placed in xy-plane as shown. The magnitude and direction of current is shown in each figure. Column-I gives statements regarding the x and y components of magnetic field at a point P whose coordinates are P(0, 0, d). Match the statements in column-I with the corresponding figures in column-II.

	Column-I		**Column-II**
(A)	The x component of magnetic field at point P is zero in	(P)	
(B)	The z component of magnetic field at point P is zero in	(Q)	
(C)	The magnitude of magnetic field at point P is $\frac{\mu_0 i}{4\pi d}$ in	(R)	
(D)	The magnitude of magnetic field at point P is less than $\frac{\mu_0 i}{2\pi d}$ in	(S)	

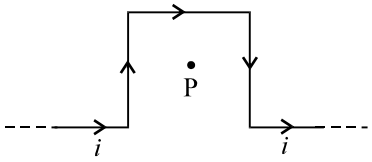
EXERCISE-II

1. Figure shows a straight wire of length *l* carrying a current *i*. Find the magnitude of magnetic field produced by the current at point P.



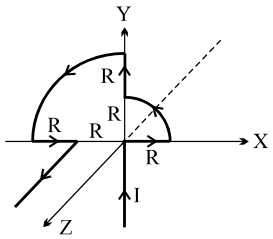
2. Two circular coils A and B of radius $\frac{5}{\sqrt{2}}$ cm and 5 cm respectively carry current 5 Amp and $\frac{5}{\sqrt{2}}$ Amp respectively. The plane of B is perpendicular to plane of A and their centres coincide. Find the magnetic field at the centre.

3. Find the magnetic field at the centre P of square of side a shown in figure.

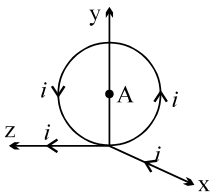


4. There is a constant homogeneous electric field of 200 V m^{-1} within the region x = 0 and x = 0.167 m pointing in the positive x-direction. There is a constant homogeneous magnetic field B within the region x = 0.167 m & x = 0.334 m pointing in the z-direction. A proton at rest at the origin (x = 0 , y = 0) is released. Find the strength of the magnetic field (in mT) B, so that the proton is deflected back to just reach x = 0, y = –0.167m. (mass of the proton = 1.67 × 10^{-27} kg)

5. Find the magnetic induction at the origin in the figure shown.

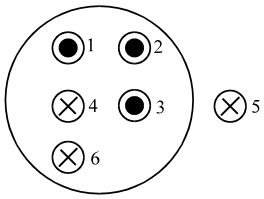


6. Find the magnitude of the magnetic induction B of a magnetic field generated by a system of thin conductors along which a current *i* is flowing at a point A (O, R, O), that is the centre of a circular conductor of radius R. The ring is in yz plane.



7. Two circular coils of wire each having a radius of 4 cm and 10 turns have a common axis and are 6 cm apart. If a current of 1 A passes through each coil in the opposite direction find the magnetic induction.
(i) At the centre of either coil (ii) At a point on the axis, midway between them.

8. Six wires of current I_1 = 1A, I_2 = 2A, I_3 = 3A, I_4 = 1A, I_5 = 5A and I_6 = 4A cut the page perpendicularly at the points 1, 2, 3, 4, 5 and 6 respectively as shown in the figure. Find the value of the integral $\oint \vec{B}.d\vec{l}$ around the closed path.



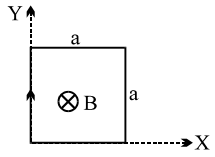
9. Electric charge *q* is uniformly distributed over a rod of length *l*. The rod is placed parallel to a long wire carrying a current *i*. The separation between the rod and the wire is a. Find the force needed to move the rod along its length with a uniform velocity *v*.

10. An electron moving with a velocity $5 \times 10^6 ms^{-1} \hat{i}$ in the uniform electric field of $5 \times 10^7 Vm^{-1} \hat{j}$. Find the magnitude and direction of a minimum uniform magnetic field in tesla that will cause the electron to move undeviated along its original path.

11. A charged particle (charge q, mass m) has velocity v_0 at origin in +x direction. In space there is a uniform magnetic field B in –z direction. Find the y coordinate of particle when is crosses y axis.

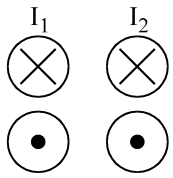
12. A conducting circular loop of radius r carries a constant current i. It is placed in a uniform magnetic field $\vec{B}_0$ such that $\vec{B}_0$ is perpendicular to the plane of the loop. Find the magnetic force acting on the loop is

13. A rectangular loop of wire is oriented with the left corner at the origin, one edge along X-axis and the other edge along Y-axis as shown in the figure. A magnetic field is into the page and has a magnitude that is given by $\beta = \alpha y$ where α is contant. Find the total magnetic force on the loop if it carries current i.

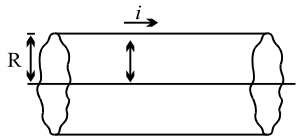


14. A particle of charge +q and mass m moving under the influence of a uniform electric field $E\hat{i}$ and a magnetic field $B\hat{k}$ enters in I quadrant of a coordinate system at a point (0, a) with initial velocity $v\hat{i}$ and leaves the quadrant at a point (2a, 0) with velocity $-2v\hat{j}$. Find
(a) Magnitude of electric field
(b) Rate of work done by the electric field at point (0, a)
(c) Rate of work done by both the fields at (2a, 0).

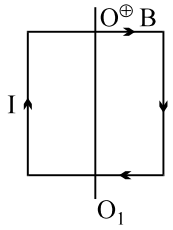
15. A system of long four parallel conductors whose sections with the plane of the drawing lie at the vertices of a square there flow four equal currents. The directions of these currents are as follows : those marked $\otimes$ point away from the reader, while those marked with a dot point towards the reader. How is the vector of magnetic induction directed at the centre of the square?



16. A cylindrical conductor of radius R carries a current along its length . The current density J, however, it is not uniform over the cross section of the conductor but is a function of the radius according to J = br, where b is a constant. Find an expression for the magnetic field B. (a) at $r_1 < R$ & (b) at distance $r_2 > R$, mesured from the axis

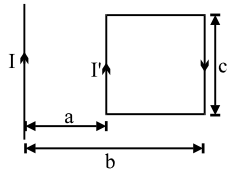


17. A square current carrying loop made of thin wire and having a mass m = 10 g can rotate without friction with respect to the vertical axis OO_1, passing through the centre of the loop at right angles to two opposite sides of the loop. The loop is placed in a homogeneous magnetic field with an induction $B = 10^{-1}$ T directed at right angles to the plane of the drawing. A current I = 2A is flowing in the loop. Find the period of small oscillations that the loop performs about its position of stable equilibrium.

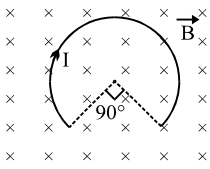


18. A proton beam passes without deviation through a region of space where there are uniform transverse mutually perpendicular electric and magnetic field with E and B. Then the beam strikes a grounded target. Find the force imparted by the beam on the target if the beam current is equal to I.

19. An infinitely long straight wire carries a conventional current I as shown in the figure. The rectangular loop carries a conventional current I' in the clockwise direction. Find the net force on the rectangular loop.

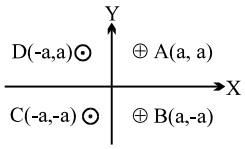


20. An arc of a circular loop of radius R is kept in the horizontal plane and a constant magnetic field B is applied in the vertical direction as shown in the figure. If the arc carries current I then find the force on the arc.

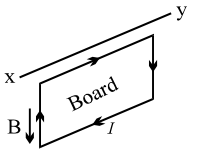


21. Four long wires each carrying current I as shown in the figure are placed at the points A, B, C and D. Find the magnitude and direction of

(i) magnetic field at the centre of the square. (ii) force per metre acting on wire at point D.



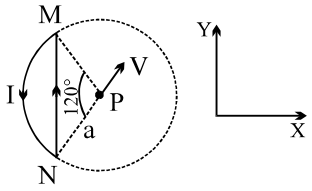
22. A square cardboard of side *l* and mass *m* is suspended from a horizontal axis XY as shown in figure. A single wire is wound along the periphery of board and carrying a clockwise current I. At t = 0, a vertical downward magnetic field of induction B is switched on. Find the minimum value of B so that the board will be able to rotate up to horizontal level.



23. A wire loop carrying current I is placed in the X-Y plane as shown in the figure

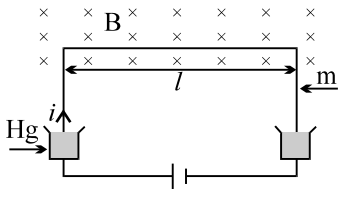
(a) If a particle with charge +Q and mass m is placed at the centre P and given a velocity along NP (fig). Find its instantaneous acceleration

(b) If an external uniform magnetic induction field $\vec{B} = B\hat{i}$ is applied, find the torque acting on the loop due to the field.



24. A U-shaped wire of mass m and length *l* is immersed with its two ends in mercury (see figure). The wire is in a homogeneous field of magnetic induction **B**. If a charge, that is, a current pulse q = $\int i dt$, is sent through the wire, the wire will jump up.

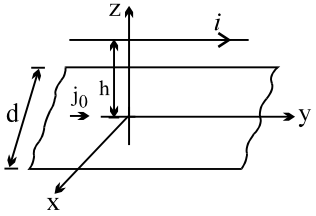
Calculate, from the height h that the wire reaches, the size of the charge or current pulse, assuming that the time of the current pulse is very small in comparision with the time of flight. Make use of the fact that impulse of force equals $\int F\,dt$, which equals mv. Evaluate q for B = 0.1 Wb/m^2, m = 10 gm, ℓ = 20 cm & h = 3 meters. [g = 10 m/s^2]



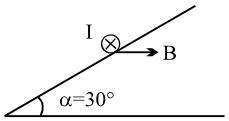
25. (a) A rigid circular loop of radius r & mass m lies in the xy plane on a flat table and has a current I flowing in it. At this particular place , the earth's magnetic field is $\vec{B} = \vec{B}_x\,\hat{i} + \vec{B}_y\,\hat{j}$. How large must I be before one edge of the loop will lift from table?

(b) Repeat if, $\vec{B} = \vec{B}_x\,\hat{i} + \vec{B}_z\,\hat{k}$.

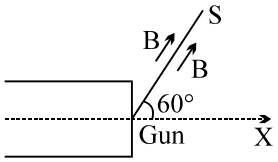
26. A conductor carrying a current i is placed parallel to a current per unit width j_0 and width d, as shown in the figure. Find the force per unit lenght on the coductor.



27. The figure shows a conductor of weight 1.0 N and length L = 0.5 m placed on a rough inclined plane making an angle 30^0 with the horizontal so that conductor is perpendicular to a uniform horizontal magnetic field of induction B = 0.10 T. The coefficient of static friction between the conductor and the plane is 0.1. A current of I = 10 A flows through the conductor inside the plane of this paper as shown. What is the force needed to be the applied parallel to the inclined plane to sustaining the conductor at rest?



28. An electron gun G emits electron of energy 2kev traveling in the (+)ve x-direction. The electron are required to hit the spot S where GS = 0.1m & the line GS makes an angle of 60° with the x-axis, as shown in the fig. A uniform magnetic field $\vec{B}$ parallel to GS exists in the region outsiees to electron gun. Find the minimum value of B needed to make the electron hit S.

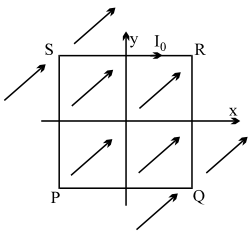


29. Two coils each of 100 turns are held such that one lies in the vertical plane with their centres coinciding. The radius of the vertical coil is 20 cm and that of the horizontal coil is 30 cm. How would you neutralize the magnetic field of the earth at their common centre? What is the current to be passed through each coil? Horizontal component of earth's magnetic induction = 3.49×10^{-5} T and angle of dip = 30º.

30. A uniform, constant magnetic field $\vec{B}$ is directed at an angle of 45º to the x-axis in the xy-plane, PQRS is a rigid square wire frame carrying a steady current I_0(clockwise), with its centre at the origin O. At time t = 0, the frame is at rest in the position shown in the figure, with its sides parallel to the x & y axes. Each side of the frame is of mass M & length L.

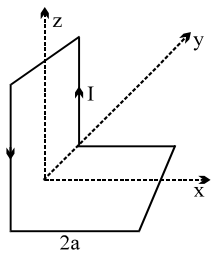
(a) What is the torque $\vec{\tau}$ about O acting on the frame due to the magnetic field?

(b) Find the angle by which the frame rotates under the action of this torque in a short interval of time Δt, & the axis about which this rotation occurs (Δt is so short that any variation in the torque during this interval may be neglected) Given the moment of inertia of the frame about an axis through its centre perpendicular to its plane is 4/3 ML^2.



EXERCISE-III

1. A non-planar loop of conducting wire carrying a current I is placed as shown in the figure. Each of the straight sections of the loop is of length 2a. The magnetic field due to this loop at the point P (a, 0, a) points in the direction **[JEE, 2001 (Scr)]**



(A) $\frac{1}{\sqrt{2}}(-\hat{j}+\hat{k})$ (B) $\frac{1}{\sqrt{3}}(-\hat{j}+\hat{k}+\hat{i})$

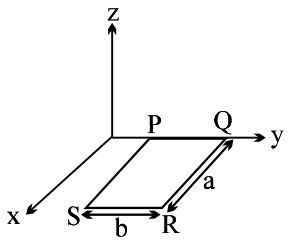
(C) $\frac{1}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$ (D) $\frac{1}{\sqrt{2}}(\hat{i}+\hat{k})$

2. A rectangular loop PQRS made from a uniform wire has length a, width b and mass m. It is free to rotate about the arm PQ, which remains hinged along a horizontal line taken as the y-axis (see figure). Take the vertically upward direction as the z-axis. A uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{k})B_0$ exists in the region. The loop is held in the x-y plane and a current I is passed through it. The loop is now released and is found to stay in the horizontal position in equilibrium.

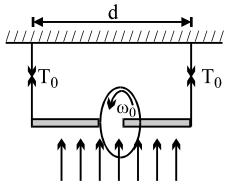
(a) What is the direction of the current I in PQ?

(b) Find the magnetic force on the arm RS.

(c) Find the expression for I in terms of B_0, a, b and m. **[JEE 2002, 1+1+3]**



3. A wheel of radius R having charge Q, uniformly distributed on the rim of the wheel is free to rotate about a light horizontal rod. The rod is suspended by light inextensible stringe and a magnetic field B is applied as shown in the figure. The initial tensions in the strings are T_0. If the breaking tension of the strings are $\frac{3T_0}{2}$, find the maximum angular velocity ω_0 with which the wheel can be rotate. **[JEE 2003]**



4. In a moving coil galvanometer, torque on the coil can be expressed as τ = ki, where i is current through the wire and k is constant. The rectangular coil of the galvanometer having numbers of turns N, area A and moment of inertia I is placed in magnetic field B. Find

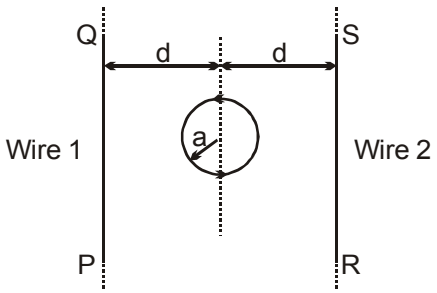
(a) k in terms of given parameters N, I, A and B.

(b) the torsional constant of the spring, if a current i_0 produces a deflection of $\pi/2$ in the coil in reaching equilibrium position.

(c) the maximum angle through which coil is deflected, if charge Q is passed through the coil almost instantaneously. (Ignore the damping in mechanical oscillations) **[JEE 2005]**

Comprehension (Q. 5 to Q. 6) **[JEE-Advanced-2014]**

The figure shows a circular loop of radius *'a'* with two long parallel wires (numbered 1 and 2) all in the plane of the paper. The distance of each wire from the centre of the loop is *d*. The loop and the wires are carrying the same current *I*. The current in the loop is in the counterclockwise direction if seen from above.



5. When $d \approx a$ but wires are not touching the loop, it is found that thenet magnetic field on the axis of the loop is zero at a height h above the loop. In that case

(A) Current in wire 1 and wire 2 is the direction PQ and RS, respectively and $h \approx a$.

(B) Current in wire 1 and wire 2 is the direction PQ and SR, respectively and $h \approx a$.

(C) Current in wire 1 and wire 2 is the direction PQ and SR, respectively and $h \approx 1.2a$.

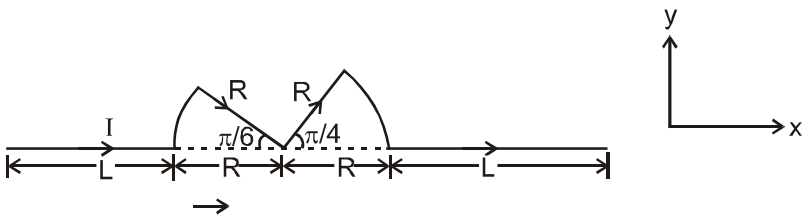
(D) Current in wire 1 and wire 2 is the direction PQ and RS, respectively and $h \approx 1.2a$.

6. Consider $d >> a$, and the loop is rotated about its diameter parallel to the wires by 30º from the position shown in the figure. If the currents in the wires are in the opposite directions, the torque on the loop at its new position will be (assume that the net field due to the wires is constant over the loop).

(A) $\frac{\mu_0 I^2 a^2}{d}$ (B) $\frac{\mu_0 I^2 a^2}{2d}$ (C) $\frac{\sqrt{3}\mu_0 I^2 a^2}{d}$ (D) $\frac{\sqrt{3}\mu_0 I^2 a^2}{2d}$

7. Two parallel wires in the plane of the paper are distance X_0 apart. A point charge is moving with speed u between the wires in the same plane at a distance X_1 from one of the wires. When the wires carry current of magnitude I in the same direction, the radius of curvature of the path of the point charge is R_1. In contrast, if the currents I in the two wires have directions opposite to each other, the radius of curvature of the path is R_2. If $\frac{X_0}{X_1} = 3$, the value of $\frac{R_1}{R_2}$ is **[JEE-Advanced-2014]**

8. A conductor (shown in the figure) carrying constant current I is kept in the x-y plane in a uniform magnetic field $\vec{B}$. If F is the magnitude of the total magnetic force acting on the conductor, then the correct statement(s) is(are) **[JEE-Advanced-2015]**



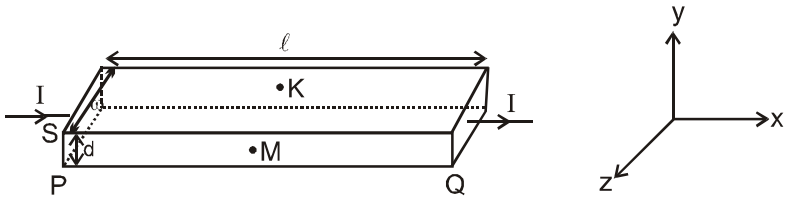
(A) If $\vec{B}$ is along $\hat{z}$, $F \propto (L + R)$ (B) If $\vec{B}$ is along $\hat{x}$, $F = 0$

(C) If $\vec{B}$ is along $\hat{y}$, $F \propto (L + R)$ (D) If $\vec{B}$ is along $\hat{z}$, $F = 0$

Paragraph for question no. 9 to 10

In a thin rectangular metallic strip a constant current I flows along the positive x-direction, as shown in the figure. The length, width and thickness of the strip are ℓ, w and d, respectively.

A unifrom magnetic field $\vec{B}$ is applied on the strip along the positive y-direction. Due to this, the charge carriers experience a net deflection along the z-direction. This results in accumulation of charge carriers on the surface PQRS and appearance of equal and opposite charges on the face opposite to PQRS. A potential difference along the z-direction is thus developed. Charge accumulation continues until the magnetic force is balanced by the electric force. The current is assumed to be uniformly distributed on the cross section of the strip and carried by electrons. **[JEE-Advanced-2015]**



9. Consider two different metallic strips (1 and 2) of the same material. Their lengths are the same, widths are ω_1 and ω_2 and thicknesses are d_1 and d_2, respectively. Two points K and M are symmetrically located on the opposite faces parallel to the x-y plane (see figure). V_1 and V_2 are the potential differences between K and M in strips 1 and 2, respectively. Then, for a given current I flowing through them in a given magnetic field strength B, the correct statement(s) is(are)

(A) If $\omega_1 = \omega_2$ and $d_1 = 2d_2$, then $V_2 = 2V_1$ (B) If $\omega_1 = \omega_2$ and $d_1 = 2d_2$, then $V_2 = V_1$

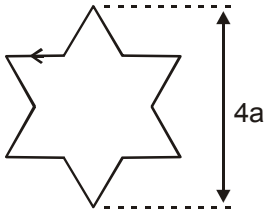
(C) If $\omega_1 = 2\omega_2$ and $d_1 = d_2$, then $V_2 = 2V_1$ (D) If $\omega_1 = 2\omega_2$ and $d_1 = d_2$, then $V_2 = V_1$

10. Consider two different metallic strips (1 and 2) of same dimensions (length ℓ, width ω and thickness d) with carrier densities n_1 and n_2, respectively. Strip 1 is placed in magnetic field B_1 and strip 2 is placed in magnetic field B_2, both along positive y-directions. Then V_1 and V_2 are the potential differences developed between K and M in strips 1 and 2, respectively. Assuming that the current I is the same for both the strips, the correct option(s) is(are)

(A) If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = 2V_1$ (B) If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = V_1$

(C) If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = 0.5\ V_1$ (D) If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = V_1$

11. A symmetric star shaped conducting wire loop is carrying a steady state current I as shown in the figure. The distance between the diametrically opposite vertices of the star is 4a. The magnitude of the magnetic field at the center of the loop is : **[JEE-Advanced-2017]**



(A) $\frac{\mu_0 I}{4\pi a} 3\left[\sqrt{3}-1\right]$ (B) $\frac{\mu_0 I}{4\pi a} 6\left[\sqrt{3}-1\right]$ (C) $\frac{\mu_0 I}{4\pi a} 6\left[\sqrt{3}+1\right]$ (D) $\frac{\mu_0 I}{4\pi a} 3\left[2-\sqrt{3}\right]$

12. Two infinitely long straight wires lie in the xy-plane along the lines x = ±R. The wire located at x = +R carries a constant current I_1 and the wire located at x = –R carries a constant current I_2. A circular loop of radius R is suspended with its centre at $(0,0\sqrt{3}R)$ and in a plane parallel to the xy-plane. This loop carries a constant current I in the clockwise direction as seen from above the loop. The current in the wire is taken to be positive if it is in the $+\hat{j}$ direction. Which of the following statements regarding the magnetic field $\vec{B}$ is(are) true? **[JEE-Advanced-2018]**

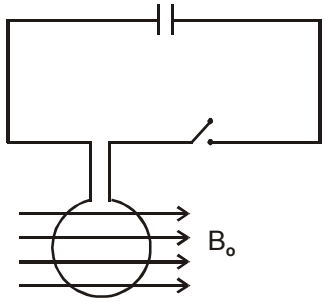
(A) If $I_1 = I_2$, then $\vec{B}$ cannot be equal to zero at the origin (0, 0, 0)

(B) If $I_1 > 0$ and $I_2 < 0$, then $\vec{B}$ can be equal to zero at the origin (0, 0, 0)

(C) If $I_1 < 0$ and $I_2 > 0$, then $\vec{B}$ can be equal to zero at the origin (0, 0, 0)

(D) If $I_1 = I_2$, then the z-component of the magnetic field at the centre of the loop is $\left(-\frac{\mu_0 I}{2R}\right)$

13. A circular coil of radius R and N turns has negligible resistance. As shown in the schematic figure, its two ends are connected to two wires and it is hanging by those wires with its plane being vertical. The wires are connected to a capacitor with charge Q through a switch. The coil is in a horizontal uniform magnetic field B_o parallel to the plane of the coil. When the switch is closed, the capacitor gets discharged through the coil in a very short time. By the time the capacitor is discharged fully, magnitude of the angular momentum gained by the coil will be (assume that the discharge time is so short that the coil has hardly rotated during this time) **[JEE-Advanced-2020]**



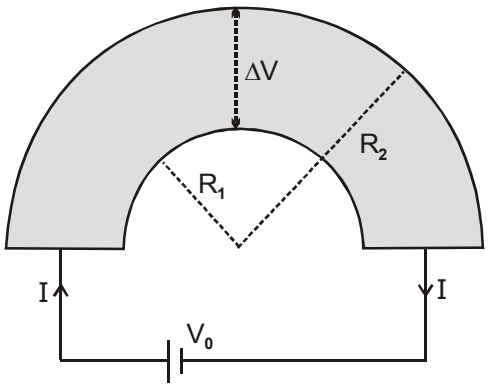
(A) $\frac{\pi}{2}NQB_oR^2$

(B) πNQB_oR^2

(C) $2\pi NQB_oR^2$

(D) $4\pi NQB_oR^2$

14. Shown in the figure is a semicircular metallic strip that has thickness t and resistivity ρ. Its inner radius is R_1 and outer radius is R_2. If a voltage V_0 is applied between its two ends, a current I flows in it. In addition, it is observed that a transverse voltage ΔV develops between its inner and outer surfaces due to purely kinetic effects of moving electrons (ignore any role of the magnetic field due to the current). Then (figure is schematic and not drawn to scale) **[JEE-Advanced-2020]**



(A) $I = \frac{V_0 t}{\pi\rho}\ln\left(\frac{R_2}{R_1}\right)$

(B) the outer surface is at a higher voltage than the inner surface

(C) the outer surface is at a lower voltage than the inner surface

(D) $\Delta V \propto I^2$

15. A light beam is described by $E = 800\sin\omega\left(t - \frac{x}{c}\right)$. An electron is allowed to move normal to the propagation of light beam with a speed of 3×10^7 ms^{-1}. What is the maximum magnetic force exerted on the electron? **[JEE-Main (Online)-2021]**

(A) 1.28×10^{-21} N (B) 12.8×10^{-17} N (C) 1.28×10^{-18} N (D) 12.8×10^{-18} N

16. A coil having N turns is wound tightly in the form of a spiral with inner and outer radii 'a' & 'b' respectively. Find the magnetic field at centre, when a current I passes through coil : **[JEE-Main (Online)-2021]**

(A) $\frac{\mu_0 I}{4(a-b)}\left[\frac{1}{a}-\frac{1}{b}\right]$ (B) $\frac{\mu_0 I}{8}\left(\frac{a-b}{a+b}\right)$ (C) $\frac{\mu_0 I}{8}\left[\frac{a+b}{a-b}\right]$ (D) $\frac{\mu_0 IN}{2(b-a)}\log_e\left(\frac{b}{a}\right)$

17. The magnetic field vector of an electromagnetic wave is given by $B = B_0\frac{\hat{i}+\hat{j}}{\sqrt{2}}\cos(kz - \omega t)$; where $\hat{i}, \hat{j}$ represents unit vector along x and y-axis respectively. At t = 0 s, two electric charges q_1 of 4π coulomb and q_2 of 2π coulomb located at $\left(0,0,\frac{\pi}{k}\right)$ and $\left(0,0,\frac{3\pi}{k}\right)$, respectively have the same velocity of $0.5\,c\,\hat{i}$, (where c is the velocity of light). The ratio of the force acting on charge q_1 to q_2 is : **[JEE-Main (Online)-2021]**

(A) $1:\sqrt{2}$ (B) 2 : 1 (C) $2\sqrt{2}:1$ (D) $\sqrt{2}:1$

18. A current of 1.5 A is flowing through a triangle, of side 9 cm each. The magnetic field at the centroid of the triangle is : (Assume that the current is flowing in the clockwise direction.) **[JEE-Main (Online)-2021]**

(A) 3×10^{-7} T, outside the plane of triangle (B) 3×10^{-5} T, inside the plane of triangle

(C) $2\sqrt{3}\times10^{-7}$ T, outside the plane of triangle (D) $2\sqrt{3}\times10^{-5}$ T, inside the plane of triangle

19. A coaxial cable consists of an inner wire of radius 'a' surrounded by an outer shell of inner and outer radii 'b' and 'c' respectively. The inner wire carries an electric current i_0, which is distributed uniformly across cross-sectional area. The outer shell carries an equal current in opposite direction and distributed uniformly. What will be the ratio of the magnetic field at a distance x from the axis when (i) x < a < x < b? **[JEE-Main (Online)-2021]**

(A) $\frac{x^2}{a^2}$ (B) $\frac{b^2-a^2}{x^2}$ (C) $\frac{x^2}{b^2-a^2}$ (D) $\frac{a^2}{x^2}$

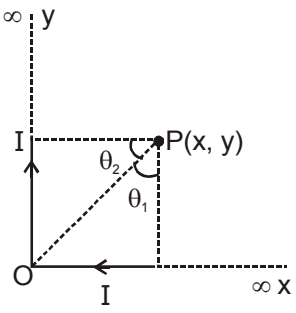
20. There are two infinitely long straight current carrying conductors and they are held at right angles to each other so that their common ends meet at the origin as shown in the figure given below. The ratio of current in both conductors is 1 : 1. The magnetic field at point P is _____. **[JEE-Main (Online)-2021]**

(A) $\frac{\mu_0 Ixy}{4\pi}\left[\sqrt{x^2+y^2}-(x+y)\right]$

(B) $\frac{\mu_0 I}{4\pi xy}\left[\sqrt{x^2+y^2}-(x+y)\right]$

(C) $\frac{\mu_0 I}{4\pi xy}\left[\sqrt{x^2+y^2}+(x+y)\right]$

(D) $\frac{\mu_0 Ixy}{4\pi}\left[\sqrt{x^2+y^2}+(x+y)\right]$



21. The fractional change in the magnetic field intensity at a distance 'r' from centre on the axis of current carrying coil of radius 'a' to the magnetic field intensity at the centre of the same coil is : (Take r < a).

[JEE-Main (Online)-2021]

(A) $\frac{3}{2}\frac{r^2}{a^2}$ (B) $\frac{2}{3}\frac{a^2}{r^2}$ (C) $\frac{3}{2}\frac{a^2}{r^2}$ (D) $\frac{2}{3}\frac{r^2}{a^2}$

22. Magnetic field at two points on the axis of a circular coil at a distance of 0.05 m and 0.2 m from the centre are in the ratio 8 : 1. The radius of coil is _____ **[JEE-Main (Online)-2021]**

(A) 0.1 m (B) 1.0 m (C) 0.2 m (D) 0.15 m

23. A proton, a deuteron and an α particle are moving with same momentum in a uniform magnetic field. The ratio of magnetic forces acting on them is _____ and their speed is ____ in the ratio.

(A) 1 : 2 : 4 and 1 : 1 : 2 (B) 1 : 2 : 4 and 2 : 1 : 1 **[JEE-Main (Online)-2021]**

(C) 4 : 2 : 1 and 2 : 1 : 1 (D) 2 : 1 : 1 and 4 : 2 : 1

24. Figure A and B show two long straight wires of circular cross-section (a and b with a < b), carrying current I which is uniformly distributed across the cross-section. The magnitude of magnetic field B varies with radius r and can be represented as : **[JEE-Main (Online)-2021]**

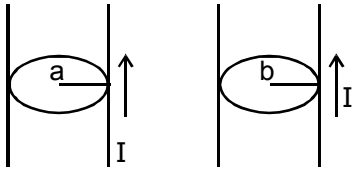
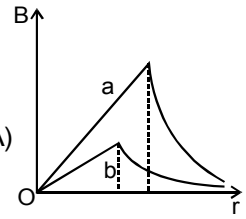
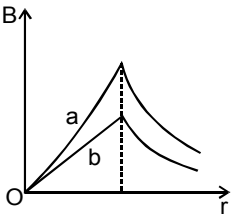
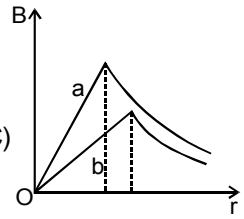
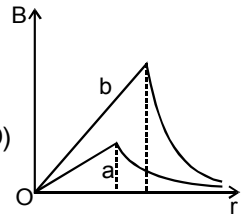


Fig. A Fig. B

(A) 

(B) 

(C) 

(D) 

25. Match List I with List II. **[JEE-Main (Online)-2021]**

	List-I		**List-II**
(a)	Capacitance, C	(i)	$M^1 L^1 T^{-3} A^{-1}$
(b)	Permittivity of free space, ε_0	(ii)	$M^{-1} L^{-3} T^4 A^2$
(c)	Permeability of free space, μ_0	(iii)	$M^{-1} L^{-2} T^4 A^2$
(d)	Electric field , E	(iv)	$M^1 L^1 T^{-2} A^{-2}$

Choose the correct answer from the options given below :

(A) (a) → (iv), (b) → (ii), (c) → (iii), (d) → (i) (B) (a) → (iii), (b) → (iv), (c) → (ii), (d) → (i)

(C) (a) → (iii), (b) → (ii), (c) → (iv), (d) → (i) (D) (a) → (iv), (b) → (iii), (c) → (ii), (d) → (i)

26. A charge Q is moving $\vec{dl}$ distance in the magnetic field $\vec{B}$. Find the value of work done by $\vec{B}$.

[JEE-Main (Online)-2021]

(A) 1 (B) Infinite (C) Zero (D) –1

27. A loop of flexible wire of irregular shape carrying current is placed in an external magentic field. Identify the effect of the field on the wire. **[JEE-Main (Online)-2021]**

(A) loop assumes circular shape with its plane normal to the field

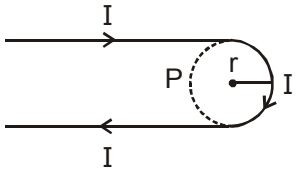
(B) loop assumes circular shape with its plane parallel to the field

(C) wire gets stretched to become straight

(D) shape of the loop remains unchanged

28. A hairpin like shape as shown in figure is made by bending a long current carrying wire. What is the magnitude of a magnetic field at point P which lies on the centre of the semicircle?

[JEE-Main (Online)-2021]



(A) $\frac{\mu_0 I}{4\pi r}(2-\pi)$ (B) $\frac{\mu_0 I}{4\pi r}(2+\pi)$ (C) $\frac{\mu_0 I}{2\pi r}(2+\pi)$ (D) $\frac{\mu_0 I}{2\pi r}(2-\pi)$

29. A solenoid of 1000 turns per metre has a core with relative permeability 500. Insulated windings of the solenoid carry an electric current of 5 A. The magnetic flux density produced by the solenoid is : (permeability of free space = $4\pi \times 10^{-7}$ H/m)

[JEE-Main (Online)-2021]

(A) π T (B) $2 \times 10^{-3}\,\pi$ T (C) $\frac{\pi}{5}$ T (D) $10^{-4}\,\pi$ T

30. A proton and an α-particle, having kinetic energy K_p and K_α respectively, enter into a magnetic field at right angles. The ratio of the radii of trajectory of proton to that of α-particle is 2 : 1. The ratio of $K_p : K_\alpha$ is.

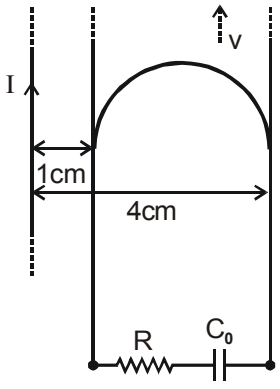
[JEE-Main (Online)-2021]

(A) 1 : 8 (B) 8 : 1 (C) 1 : 4 (D) 4 : 1

31. A long straight wire carries a current, I = 2 ampere. A semi-circular conducting rod is placed beside it on two conducting parallel rails of negligible resistance. Both the rails are parallel to the wire. The wire, the rod and the rails lie in the same horizontal plane, as shown in the figure. Two ends of the semi-circular rod are at distances 1 cm and 4 cm from the wire. At time t = 0, the rod starts moving on the rails with a speed v = 3.0 m/s (see the figure).

A resistor R = 1.4 Ω and a capacitor C_0 = 5.0 μF are connected in series between the rails. At time t = 0, C_0 is uncharged. Which of the following statement(s) is(are) correct?

[$\mu_0 = 4\pi \times 10^{-7}$ SI units. Take ln 2 = 0.7] **[JEE-Advanced-2021]**



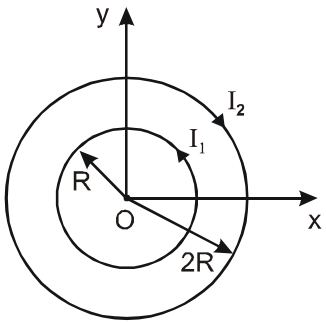
(A) Maximum current through R is 1.2×10^{-6} ampere

(B) Maximum current through R is 3.8×10^{-6} ampere

(C) Maximum charge on capacitor C_0 is 8.4×10^{-12} coulomb

(D) Maximum charge on capacitor C_0 is 2.4×10^{-12} coulomb

32. Two concentric circular loops, one of radius R and the other of radius 2R, lie in the xy-plane with the origin as their common center, as shown in the figure. The smaller loop carries current I_1 in the anti-clockwise direction and the larger loop carries current I_2 in the clockwise direction, with $I_2 > 2I_1$. $\vec{B}(x, y)$ denotes the magnetic field at a point (x, y) in the xy-plane. Which of the following statement(s) is(are) correct?



(A) $\vec{B}$ (x, y) is perpendicular to the xy-plane at any point in the plane **[JEE-Advanced-2021]**

(B) $|\vec{B}$ (x, y)| depends on x and y only through the radial distance r = $\sqrt{x^2 + y^2}$

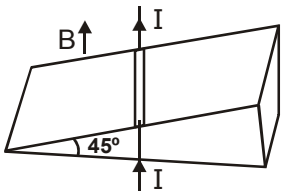
(C) $|\vec{B}$ (x, y)| is non-zero at all points for r < R

(D) $\vec{B}$ (x, y) points normally outward from the xy-plane for all the points between the two loops

33. Two charged particles, having same kinetic energy, are allowed to pass through a uniform magnetic field perpendicular to the direction of motion. If the ratio of radii of their circular paths is 6 : 5 and their respective masses ratio is 9 : 4. Then, the ratio of their charges will be **[JEE-Main (Online)-2022]**

(A) 8 : 5 (B) 5 : 4 (C) 5 : 3 (D) 8 : 7

34. As shown in the figure, a metallic rod of linear density 0.45 kg m^{-1} is lying horizontally on a smooth incline plane which makes an angle of 45° with the horizontal. The minimum current flowing in the rod required to keep it stationary, when 0.15 T magnetic field is acting on it in the vertical upward direction, will be :
{Use g = 10 m/s^2} **[JEE-Main (Online)-2022]**



(A) 30 A (B) 15 A (C) 10 A (D) 3 A

35. B_X and B_Y are the magnetic field at the centre of two coils of two coils X and Y respectively, each carrying equal current. If coil X has 200 turns and 20 cm radius and coil Y has 400 turns and 20 cm radius, the ratio of B_X and B_Y is **[JEE-Main (Online)-2022]**

(A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 4 : 1

36. A charge particle is moving in a uniform magnetic field $(2\hat{i} + 3\hat{j})$T . If it has an acceleration of $(\alpha\hat{i} - 4\hat{j})$m / s^2, then the value of α will be **[JEE-Main (Online)-2022]**

(A) 3 (B) 6 (C) 12 (D) 2

37. A cyclotron is used to accelerate proptons. If the operating magnetic field is 1.0 T and the radius of the cyclotron 'dees' is 60 cm, the kinetic energy of the accelerated protons in MeV will be
[Use $m_p = 1.6 \times 10^{-27}$ kg, $e = 1.6 \times 10^{-19}$ C] **[JEE-Main (Online)-2022]**

(A) 12 (B) 18 (C) 16 (D) 32

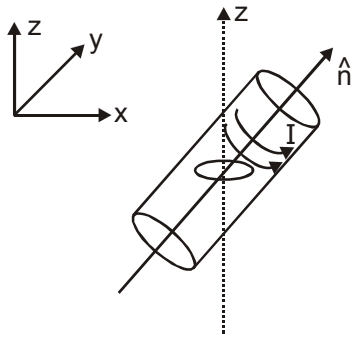
38. The magnetic field at the centre of a circular coil of radius r, due to current I flowing through it, is B. The magnetic field at a point along the axis at a distance $\frac{r}{2}$ from the cenre is **[JEE-Main (Online)-2022]**

(A) $\frac{B}{2}$ (B) 2B (C) $\left(\frac{2}{\sqrt{5}}\right)^3 B$ (D) $\left(\frac{2}{\sqrt{3}}\right)^3 B$

39. A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of magnetic field will be equal to **[JEE-Main (Online)-2022]**

(A) B (B) 2B (C) 4B (D) $\frac{B}{2}$

40. A small circular loop of area A and resistance R is fixed on a horizontal xy-plane with the center of the loop always on the axis $\hat{n}$ of a long solenoid. The solenoid has m turns per unit length and carries current I counterclockwise as shown in the figure. The magnetic field due to the solenoid is in $\hat{n}$ direction. List-I gives time dependences of $\hat{n}$ in terms of a constant angular frequency ω. List-II gives the torques experienced by the circular loop at time $t = \frac{\pi}{6\omega}$. Let $\alpha = \frac{A^2\mu_0^2 m^2 \ell^2 \omega}{2R}$. **[JEE-Advanced-2022]**



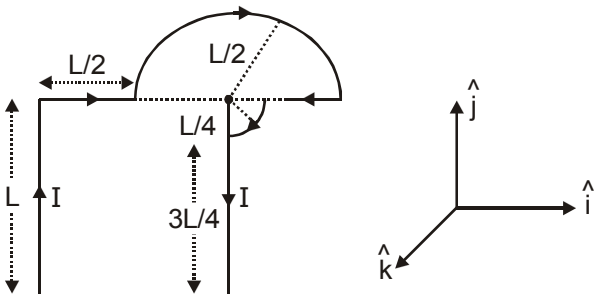
	List-I		List-II
(I)	$\frac{1}{\sqrt{2}}(\sin\omega t\,\hat{j} + \cos\omega t\,\hat{k})$	(P)	0
(II)	$\frac{1}{\sqrt{2}}(\sin\omega t\,\hat{i} + \cos\omega t\,\hat{j})$	(Q)	$-\frac{\alpha}{4}\hat{i}$
(III)	$\frac{1}{\sqrt{2}}(\sin\omega t\,\hat{i} + \cos\omega t\,\hat{k})$	(R)	$\frac{3\alpha}{4}\hat{i}$
(IV)	$\frac{1}{\sqrt{2}}(\cos\omega t\,\hat{j} + \sin\omega t\,\hat{k})$	(S)	$\frac{\alpha}{4}\hat{j}$
		(T)	$-\frac{3\alpha}{4}\hat{i}$

Which one of the following options is correct?

(A) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (T) (B) (I) $\rightarrow$ (S); (II) $\rightarrow$ (T); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (P)

(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (R) (D) (I) $\rightarrow$ (T); (II) $\rightarrow$ (Q); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

41. Which one of the following options represents the magnetic field $\vec{B}$ at O due to the current flowing in the given wire segments lying on the xy plane? **[JEE-Advanced-2022]**



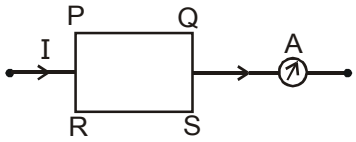
(A) $\vec{B} = \frac{-\mu_0 I}{L}\left(\frac{3}{2} + \frac{1}{4\sqrt{2}\pi}\right)\hat{k}$ (B) $\vec{B} = -\frac{\mu_0 I}{L}\left(\frac{3}{2} + \frac{1}{2\sqrt{2}\pi}\right)\hat{k}$

(C) $\vec{B} = \frac{-\mu_0 I}{L}\left(1 + \frac{1}{4\sqrt{2}\pi}\right)\hat{k}$ (D) $\vec{B} = \frac{-\mu_0 I}{L}\left(1 + \frac{1}{4\pi}\right)\hat{k}$

42. The magnetic moments associated with two closely wound circular coils A and B of radius r_A = 10 cm and r_B = 20 cm respectively are equal if : (Where N_A, I_A and N_B, I_B are number of turn and current of A and B respectively) **[JEE-Main (Online)-2023]**

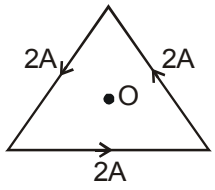
(A) $N_A I_A = 4N_B I_B$ (B) $2N_A I_A = N_B I_B$ (C) $4N_A I_A = N_B I_B$ (D) $N_A = 2N_B$

43. A current carrying rectangular loop PQRS is made of uniform wire. The length PR = QS = 5 cm and PQ = RS = 100 cm. If ammeter current reading changes from I to 2I, the ratio of magnetic forces per unit length on the wire PQ due to wire RS in the two cases respectively $\left(f_{PQ}^{I} : f_{PQ}^{2I}\right)$ is : **[JEE-Main (Online)-2023]**



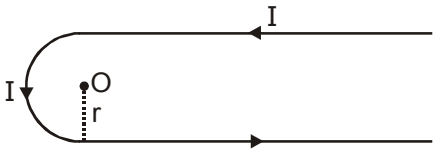
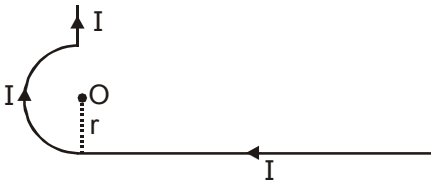
(A) 1 : 2 (B) 1 : 3 (C) 1 : 4 (D) 1 : 5

44. As shown in the figure, a current of 2A flowing in an equilateral triangle of side $4\sqrt{3}$ cm. The magnetic field at the centroid O of the triangle is : (Neglect the effect of earth's magnetic field) **[JEE-Main (Online)-2023]**



(A) $4\sqrt{3} \times 10^{-5}$ T (B) $4\sqrt{3} \times 10^{-4}$ T (C) $3\sqrt{3} \times 10^{-5}$ T (D) $\sqrt{3} \times 10^{-4}$ T

45. Match **List-I** with **List-II**

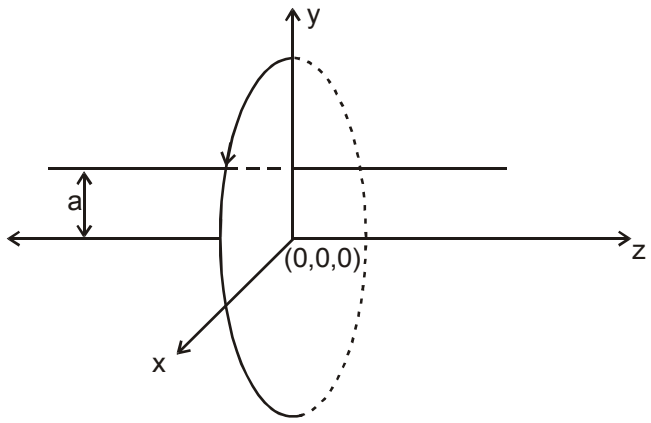
List-I (Current configuration)		**List-II (Magnitude of Magnetic Field at point O)**	
(A)		(I)	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 2]$
(B)		(II)	$B_0 = \frac{\mu_0}{4}\frac{I}{r}$
(C)		(III)	$B_0 = \frac{\mu_0 I}{2\pi r}[\pi - 1]$
(D)		(IV)	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 1]$

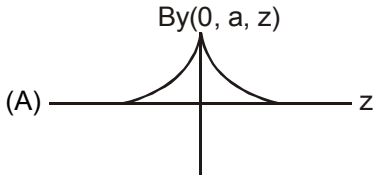
Choose the correct answer from the options given below : **[JEE-Main (Online)-2023]**

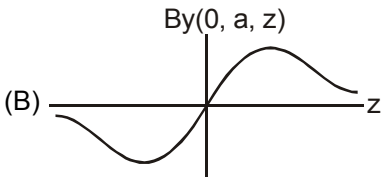
(A) A-III B-I C-IV D-II (B) A-III B-IV C-I D-II

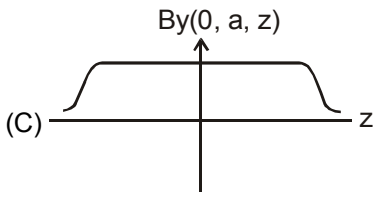
(C) A-II B-I C-IV D-III (D) A-I B-III C-IV D-II

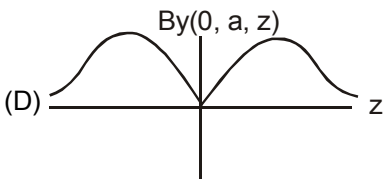
46. A single current carrying loop of wire carrying current I flowing in anticlokwise direction seen from +ve z direction and lying in xy plane is shown in figure. The plot of $\hat{j}$ component of magnetic field (By) at a distance 'a' (less than radius of the coil) & on yz plane vs z coordinate looks like **[JEE-Main (Online)-2023]**



(A) 

(B) 

(C) 

(D) 

47. A long straight wire of circular cross-section (radius a) is carrying steady current I. The current I is uniformly distributed across this cross-section. The magnetic field is **[JEE-Main (Online)-2023]**

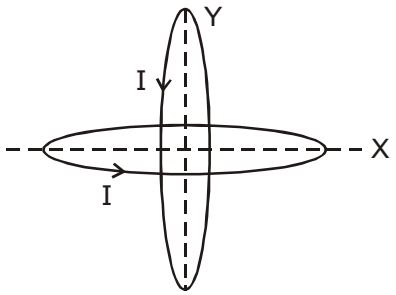
(A) directly proportional to r in the region r < a and inversely proportional to r in the region r > a

(B) inversely proportional to r in the region r < a and uniform throughout in the region r > a

(C) uniform in the region r < a and inversely proportional to distance r from the axis, in the region r > a

(D) zero in the region r < a < a and inversely proportional to r in the region r > r > a

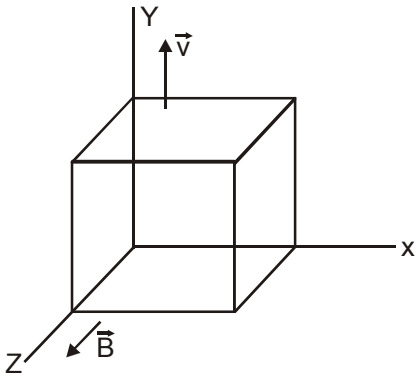
48. Two identical circular wires of radius 20 cm and carrying current $\sqrt{2}$A are placed in perpendicular planes as shown in figure. The net magnetic field at the centre of the circular wires is _____ × 10^{-8} T.



(Take : π = 3.14) **[JEE-Main (Online)-2023]**

49. A metallic cube of side 15 cm moving along y-axis at a uniform velocity of 2 ms^{-1}. In a region of uniform magnetic field of magnitude 0.5 T directed along z-axis. In equilibrium the potential difference between the faces of higher and lower potential developed because of the motion through the field will be ___ mV.

[JEE-Main (Online)-2023]



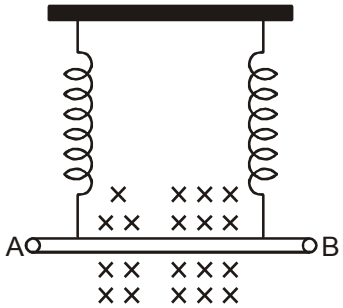
50. A proton with a kinetic energy of 2.0 eV moves into a region of uniform magnetic field of magnitude $\frac{\pi}{2}$ × 10^{-3} T. The angle between the direction of magnetic field and velocity of proton is 60°. The pitch of the helical path taken by the proton is _____ cm. **[JEE-Main (Online)-2023]**

(Take, mass of proton = 1.6 × 10^{-27} kg and Charge on proton = 1.6 × 10^{-19} C).

51. The magnetic intensity at the center of a long current carrying solenoid is found to be 1.6 × 10^3 Am^{-1}. If the number of turns is 8 per cm, then the current flowing through the solenoid is___A.

[JEE-Main (Online)-2023]

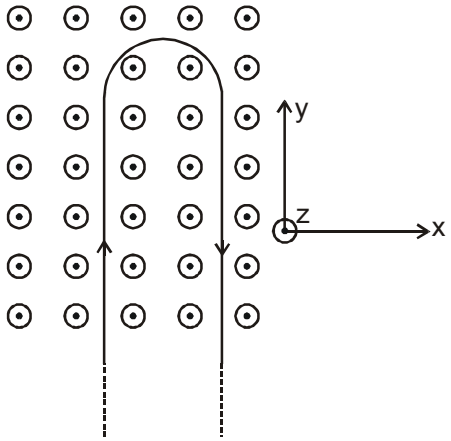
52. A straight wire AB of mass 40 g and length 50 cm is suspended by a pair of flexible leads in uniform magnetic field of magnitude 0.40 T as shown in the figure. The magnitude of the current required in the wire to remove the tension in the supporting leads is_____ A. (Take g = 10 ms^{-2}) **[JEE-Main (Online)-2023]**



53. Two particles X and Y having equal charges are being accelerated through the same potential difference. Thereafter they enter normally in a region of uniform magnetic field and describes circular paths of radii R_1 and R_2 respectively. The mass ratio of X and Y is **[JEE-Main (Online)-2024]**

(A) $\left(\frac{R_2}{R_1}\right)$ (B) $\left(\frac{R_2}{R_1}\right)^2$ (C) $\left(\frac{R_1}{R_2}\right)^2$ (D) $\left(\frac{R_1}{R_2}\right)$

54. A rigid wire consists of a semicircular portion of radius R and two straight sections. The wire is partially immerged in a perpendicular magnetic field $B = B_0 \hat{j}$ as shown in figure. The magnetic force on the wire if it has a current i is : **[JEE-Main (Online)-2024]**



(A) $2iBR\hat{j}$ (B) $-2iBR\hat{j}$ (C) $-iBR\hat{j}$ (D) $iBR\hat{j}$

55. A proton moving with a constant velocity passes through a region of space without any change in its velocity. If $\vec{E}$ and $\vec{B}$ represent the electric and magnetic fields respectively, then the region of space may have :

(a) E = 0, B = 0 (b) E = 0, B $\neq$ 0 (c) E $\neq$ 0, B = 0 (d) E $\neq$ 0, B $\neq$ 0

Choose the most appropriate answer from the options given below : **[JEE-Main (Online)-2024]**

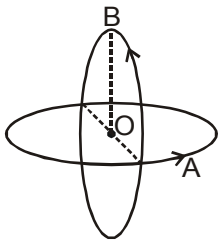
(A) (a), (c) and (d) only (B) (b), (c) and (d) only (C) (a), (b) and (d) only (D) (a), (b) and (c) only

56. The magnetic moment of a bar magnet is 0.5 Am^2. It is suspended in a uniform magnetic field of 8×10^{-2} T. The work done in rotating it from its most stable to most unstable position is **[JEE-Main (Online)-2024]**

(A) 8×10^{-2} J (B) Zero (C) 4×10^{-2} J (D) 16×10^{-2} J

57. Two insulated circular loop A and B of radius 'a' carrying a current of 'I' in the anti clockwise direction as shown in the figure. The magnitude of the magnetic induction at the centre will be

[JEE-Main (Online)-2024]



(A) $\frac{2\mu_0 I}{a}$ (B) $\frac{\mu_0 I}{\sqrt{2}a}$

(C) $\frac{\mu_0 I}{2a}$ (D) $\frac{\sqrt{2}\mu_0 I}{a}$

58. A proton and a deutron (q = +e, m = 2.0u) having same kinetic energies enter a region of uniform magnetic field $\vec{B}$, moving prependicular to $\vec{B}$. The ratio of the radius r_d of deutron path to the radius r_p of the proton path is :

[JEE-Main (Online)-2024]

(A) 1 : 1 (B) $\sqrt{2}:1$ (C) $1:\sqrt{2}$ (D) 1 : 2

59. A solenoind of length 0.5 m has a radius of 1 cm and is made up of 'm' number of turns. It carries a current of 5A. If the magnitude of the magnetic field inside the solenoid is 6.28×10^{-3} T then the value of m is_____

[JEE-Main (Online)-2024]

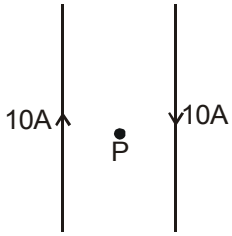
60. Two circular coils P and Q of 100 turns each have same radius of π cm. The currents in P and Q are 1A and 2A respectively. P and Q are placed with their planes mutually perpendicular with their centers coincide. The resultant magnetic field induction at the center of the coils is $\sqrt{x}$ mT, where x = _____.

[Use $\mu_0 = 4\pi \times 10^{-7}$ TmA^{-1}]

[JEE-Main (Online)-2024]

61. Two long, straight wires carry equal currents in opposite directions as shown in figure. The separation between the wires is 5.0 cm.The magnitude of the magnetic field at a point P midway between the wires is_____μT. (Given : $\mu_0 = 4\pi \times 10^{-7}$ TmA^{-1})

[JEE-Main (Online)-2024]



62. A 2A current carrying straight metal wire of resistance 1Ω, resistivity 2×10^{-6} Ωm, area of cross-section 10 mm^2 and mass 500 g is suspended horizontally in mid air by applying a uniform magnetic field $\vec{B}$. The magnitude of B is _____ $\times 10^{-1}$ T (Given : g = 10 m/s^2)

[JEE-Main (Online)-2024]

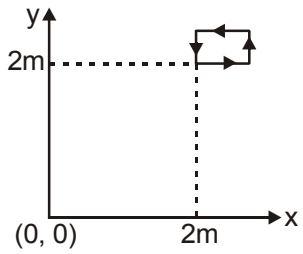
63. A square loop of edge length 2 m carrying current of 2 A is placed with its edges parallel to the x-y axis. A magnetic field is passing through the x-y plane and expressed as $\vec{B} = B_0(1+4x)\hat{k}$, where B_0 = 5 T. The net magnetic force experienced by the loop is _____ N.

[JEE-Main (Online)-2024]

64. A moving coil galvanometer has 100 turns and each turn has an area of 2.0 cm^2. The magnetic field produced by the magnet is 0.01 T and the deflection in the coil is 0.05 radian when a current of 10 mA is passed through it. The torsional constant of the suspension wire is $x \times 10^{-5}$ N-m/rad. The value of x is_____.

[JEE-Main (Online)-2024]

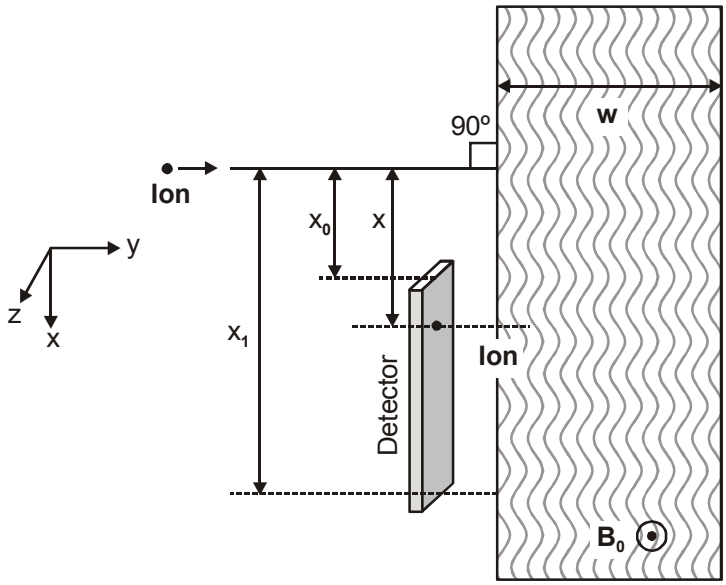
65. The magnetic field existing in a region is given by $\vec{B} = 0.2(1+2x)\hat{k}T$. A square loop of edge 50 cm carrying 0.5 A current is placed in x-y plane with its edges parallel to the x-y axis, as shown in figure. The magnitude of the net magnetic force experienced by the loop is _____mN. **[JEE-Main (Online)-2024]**



66. An infinitely long wire, located on the z-axis, carries a current I along the +z-direction and produces the magnetic field $\vec{B}$. The magnitude of the line integral $\int \vec{B} \cdot \vec{d\ell}$ along a straight line from the point $(-\sqrt{3}\,a, a, 0)$ to $(a, a, 0)$ is given by [μ_0 is the magnetic permeability of free space.] **[JEE-Advanced-2024]**

(A) $\frac{7\mu_0 I}{24}$ (B) $\frac{7\mu_0 I}{12}$ (C) $\frac{\mu_0 I}{8}$ (D) $\frac{\mu_0 I}{6}$

67. A positive, singly ionized atom of mass number A_M is accelerated from rest by the voltage 192 V. Thereafter, it enters a rectangular region of width w with magnetic field $\vec{B}_0 = 0.1\hat{k}$ Tesla, as shown in the figure. The ion finally hits a detector at the distance x below its starting trajectory.
[Given : Mass of neutron/proton = (5/3) × 10^{-27} kg, charge of the electron = 1.6 × 10^{-19} C.]



Which of the following option(s) is(are) correct **[JEE-Advanced-2024]**

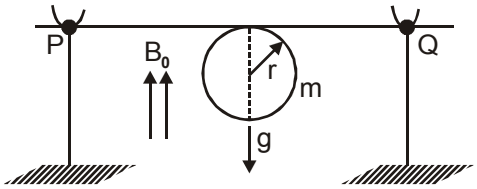
(A) The value of x for H^+ ion is 4 cm.

(B) The value of x for an ion with A_M = 144 is 48 cm.

(C) For detecting ions with $1 \le A_M \le 196$, the minimum height ($x_1 - x_0$) of the detector is 55 cm.

(D) The minimum width w of the region of the magnetic field for detecting ions with A_M = 196 is 56 cm.

68. A thin stiff insulated metal wire is bent into a circular loop with its two ends extending tangentially from the same point of the loop. The wire loop has mass m and radius r and it is in a uniform vertical magnetic field B_0, as shown in the figure. Initially, it hangs vertically downwards, because of acceleration due to gravity g, on two conducting supports at P and Q. When a current I is passed through the loop, the loop turns about the line PQ by an angle θ given by **[JEE-Advanced-2024]**



(A) $\tan\theta = \pi r I B_0 / (mg)$ (B) $\tan\theta = 2\pi r I B_0 / (mg)$
(C) $\tan\theta = \pi r I B_0 / (2mg)$ (D) $\tan\theta = mg / (\pi r I B_0)$

69. If B is magnetic field and μ_0 is permeability of free space, then the dimensions of (B/μ_0) is **[JEE-Mains Online-2025]**

(A) $MT^{-2}A^{-1}$ (B) $L^{-1}A$ (C) $LT^{-2}A^{-1}$ (D) $ML^2T^{-2}A^{-1}$

70. Match **List-I** with **List-II**. **[JEE-Mains Online-2025]**

	List-I		**List-II**
(A)	Magnetic induction	(I)	Ampere meter^2
(B)	Magnetic intensity	(II)	Weber
(C)	Magnetic flux	(III)	Gauss
(D)	Magnetic moment	(IV)	Ampere meter

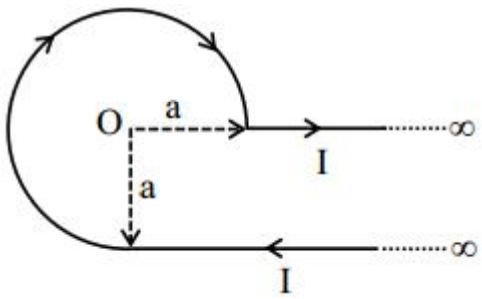
Choose the correct answer from the options given below :

(A) (A)-(III), (B)-(IV), (C)-(I), (D)-(II) (B) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
(C) (A)-(I), (B)-(II), (C)-(III), (D)-(IV) (D) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

71. Consider a long straight wire of a circular cross-section (radius a) carrying a steady current I. The current is uniformly distributed across this cross-section. The distances from the centre of the wire's cross-section at which the magnetic field [inside the wire, outside the wire] is half of the maximum possible magnetic field, any where due to the wire, will be **[JEE-Mains Online-2025]**

(A) $\left[\frac{a}{4}, \frac{3a}{2}\right]$ (B) $\left[\frac{a}{2}, 2a\right]$ (C) $\left[\frac{a}{2}, 3a\right]$ (D) $\left[\frac{a}{4}, 2a\right]$

72. An infinite wire has a circular bend of radius a, and carrying a current I as shown in figure. The magnitude of magnetic field at the origin O of the arc is given by **[JEE-Mains Online-2025]**

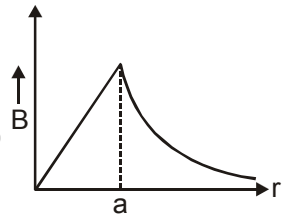


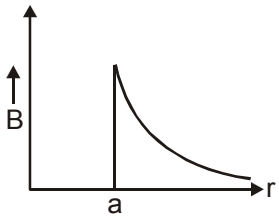
(A) $\frac{\mu_0}{4\pi}\frac{I}{a}\left[\frac{\pi}{2}+1\right]$ (B) $\frac{\mu_0}{4\pi}\frac{I}{a}\left[\frac{3\pi}{2}+1\right]$ (C) $\frac{\mu_0}{2\pi}\frac{I}{a}\left[\frac{\pi}{2}+2\right]$ (D) $\frac{\mu_0}{4\pi}\frac{I}{a}\left[\frac{3\pi}{2}+2\right]$

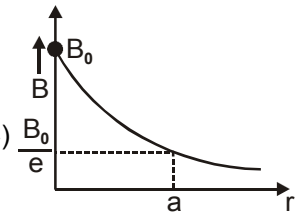
73. Consider a long thin conducting wire carrying a uniform current I. A particle having mass "M" and charge "q" is released at a distance "a" from the wire with a speed v_0 along the direction of current in the wire. The particle gets attracted to the wire due to magnetic force. The particle turns round when it is at distance x from the wire. The value of x is [μ_0 is vacuum permeability] **[JEE-Mains Online-2025]**

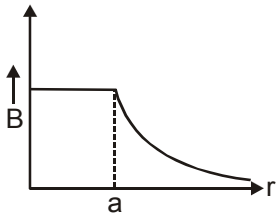
(A) $a\left[1-\frac{mv_0}{2q\mu_0 I}\right]$ (B) $\frac{a}{2}$ (C) $a\left[1-\frac{mv_0}{q\mu_0 I}\right]$ (D) $ae^{\frac{-4\pi mv_0}{q\mu_0 1}}$

74. A long straight wire of a circular cross-section with radius 'a' carries a steady current I. The current I is a uniformly distributed across this cross-section. The plot of magnitude of magnetic field B with distance r from the centre of the wire is given by **[JEE-Mains Online-2025]**

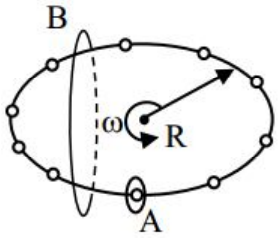
(A)


(B)


(C)


(D)


75. N equally spaced charges each of value q, are placed on a circle of radius R. The circle rotates about its axis with an angular velocity ω as shown in the figure. A bigger Amperian loop B encloses the whole circle where as a smaller Amperian loop A encloses a small segment. The difference between enclosed currents, $I_A - I_B$, for the given Amperian loops is **[JEE-Mains Online-2025]**



(A) $\frac{N^2}{2\pi}q\omega$ (B) $\frac{2\pi}{N}q\omega$ (C) $\frac{N}{2\pi}q\omega$ (D) $\frac{N}{\pi}q\omega$

76. Given below are two statements. One is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.
Assertion (A) : A electron in a certain region of uniform magnetic field is moving with constant velocity in a straight line path. **[JEE-Mains Online-2025]**
Reason (A) : The magnetic field in that region is along the direction of velocity of the electron.
In the light of the above statements, choose the **correct** answer from the options given below :
(A) (A) is false but (R) is true
(B) Both (A) and (R) are true and (R) is the correct explanation of (A)
(C) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
(D) (A) is true but (R) is false

77. Match **List-I** with **List-II**. **[JEE-Mains Online-2025]**

List-I		**List-II**
(A) Permeability of free space	(I)	$[M L^2 T^{-2}]$
(B) Magnetic field	(II)	$[M T^{-2} A^{-1}]$
(C) Magnetic moment	(III)	$[M L T^{-2} A^{-2}]$
(D) Torsional constant	(IV)	$[L^2 A]$

Choose the **correct** answer from the options given below :

(A) (A)-(I), (B)-(IV), (C)-(II), (D)-(III) (B) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

(C) (A)-(IV), (B)-(III), (C)-(I), (D)-(II) (D) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

78. Consider a moving coil galvanometer (MCG) : **[JEE-Mains Online-2025]**

A : The torsional constant in moving coil galvanometer has dimensions $[ML^2T^{-2}]$

B : Increasing the current sensitivity may not necessarily increase the voltage sensitivity.

C : If we increase number of turns (N) to its double (2N), then the voltage sensitivity doubles.

D : MCG can be converted into an ammeter by introducing a shunt resistance of large value in parallel with galvanometer.

E : Current sensitivity of MCG depends inversely on number of turns of coil.

Choose the correct answer from the options given below :

(A) A, B only (B) A, D, only (C) B, D, E only (D) A, B, E only

79. In a moving coil galvanometer, two moving coils M_1 and M_2 have the following particulars

$R_1 = 5\ \Omega$, $N_1 = 15$, $A_1 = 3.6 \times 10^{-3}\ m^2$, $B_1 = 0.25$ T

$R_2 = 7\ \Omega$, $N_2 = 21$, $A_2 = 1.8 \times 10^{-3}\ m^2$, $B_2 = 0.50$ T

Assuming that torsional constant of the springs are same for both coils, what will be the ratio of voltage sensitivity of M_1 and M_2 ? **[JEE-Mains Online-2025]**

(A) 1 : 1 (B) 1 : 4 (C) 1 : 3 (D) 1 : 2

80. Let B_1 be the magnitude of magnetic field at center of a circular coil of radius R carrying current I. Let B_2 be the magnitude of magnetic field at an axial distance 'x' from the center. For x : R = 3 : 4, $\frac{B_2}{B_1}$ is : **[JEE-Mains Online-2025]**

(A) 4 : 5 (B) 16 : 25 (C) 64 : 125 (D) 25 : 16

81. A magnetic dipole experiences a torque of $80\sqrt{3}$ Nm when placed in uniform magnetic field in such a way that dipole moment makes angle of 60° with magnetic field. The potential energy of the dipole is : **[JEE-Mains Online-2025]**

(A) 80 J (B) $-40\sqrt{3}$ J (C) –60 J (D) –80 J

82. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : If oxygen ion (O^{-2}) and Hydrogen ion (H^+) enter normal to the magnetic field with equal momentum, then the path of O^{-2} ion has a smaller curvature than that of H^+.

Reason R : A proton with same linear momentum as an electron will form a path of smaller radius of curvature on entering a uniform magnetic field perpendicularly.

In the light of the above statement, choose the correct answer from the options given below

(A) **A** is true but **R** is false

(B) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A** **[JEE-Mains Online-2025]**

(C) **A** is false but **R** is true

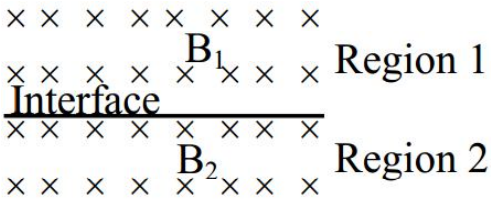
(D) Both **A** and **R** are true and **R** is the correct explanation of **A**

83. In an electromagnetic system, a quantity defined as the ratio of electric dipole moment and magnetic dipole moment has dimension of $[M^PL^QT^RA^S]$. The value of P and Q are : **[JEE-Mains Online-2025]**

(A) –1, 0 (B) –1, 1 (C) 1, –1 (D) 0, –1

84. Uniform magnetic fields of different strengths (B_1 and B_2), both normal to the plane of the paper exist as shown in the figure. A charged particle of mass m and charge q, at the interface at an instant, moves into the region 2 with velocity v and returns to the interface. It continues to move into region 1 and finally reaches the interface. What is the displacement of the particle during this movement along the interface?

[JEE-Mains Online-2025]

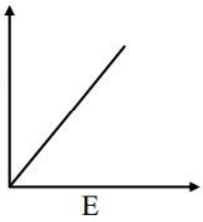


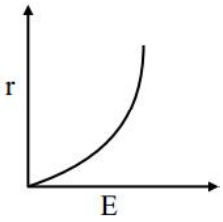
(Consider the velocity of the particle to be normal to the magnetic field and $B_2 > B_1$)

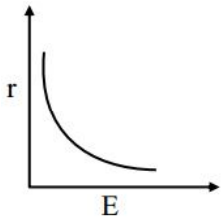
(A) $\frac{mv}{qB_1}\left(1-\frac{B_2}{B_1}\right)\times 2$ (B) $\frac{mv}{qB_1}\left(1-\frac{B_1}{B_2}\right)$ (C) $\frac{mv}{qB_1}\left(1-\frac{B_2}{B_1}\right)$ (D) $\frac{mv}{qB_1}\left(1-\frac{B_1}{B_2}\right)\times 2$

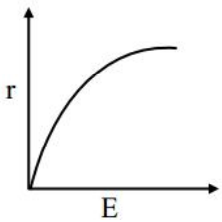
85. A particle of charge q, mass m and kinetic energy E enters in magnetic field perpendicular to its velocity and undergoes a circular arc of radius(r). Which of the following curves represents the variation of r with E?

[JEE-Mains Online-2025]

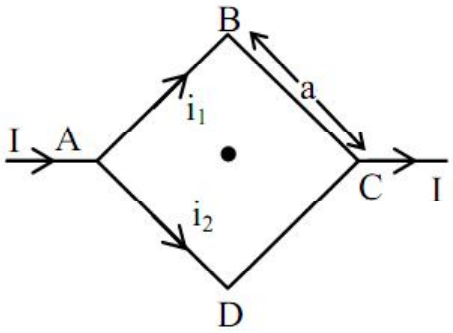
(A) 

(B) 

(C) 

(D) 

86. Figure shows a current carrying square loop ABCD of edge length is 'a' lying in a plane. If the resistance of the ABC part is r and that of ADC part is 2r, then the magnitude of the resultant magnetic field at centre of the square loop is : **[JEE-Mains Online-2025]**

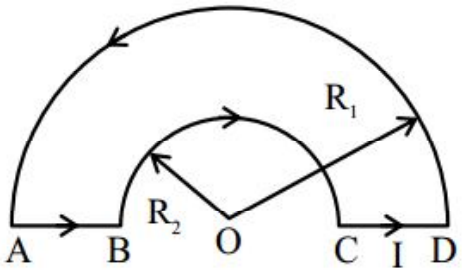


(A) $\frac{3\pi\mu_0 I}{\sqrt{2}a}$ (B) $\frac{\mu_0 I}{2\pi a}$ (C) $\frac{\sqrt{2}\mu_0 I}{3\pi a}$ (D) $\frac{2\mu_0 I}{3\pi a}$

87. A 4.0 cm long straight wire carrying a current of 8A is placed perpendicular to an uniform magnetic field of strength 0.15 T. The magnetic force on the wire is _____ mN. **[JEE-Mains Online-2025]**

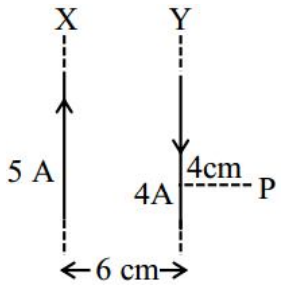
88. A particle of charge 1.6 μC and mass 16 μg is present in a strong magnetic field of 6.28 T. The particle is then fired perpendicular to magnetic field. The time required for the particle to return to original location for the first time is _____ s. (π = 3.14) **[JEE-Mains Online-2025]**

89. A loop ABCDA, carrying current I = 12 A, is placed in a plane, consists of two semi-circular segments of radius $R_1 = 6\pi$ m and $R_2 = 4\pi$ m. The magnitude of the resultant magnetic field at center O is $k \times 10^{-7}$ T. The value of k is _____ . (Given $\mu_0 = 4\pi \times 10^{-7}$ Tm A^{-1}) **[JEE-Mains Online-2025]**



90. A proton is moving undeflected in a region of crossed electric and magnetic fields at a constant speed of 2×10^5 ms^{-1}. When the electric field is switched off, the proton moves along a circular path of radius 2 cm. The magnitude of electric field is $x \times 10^4$ N/C. the value of x is _____. **[JEE-Mains Online-2025]**
Take the mass of the proton = 1.6×10^{-27} kg.

91. Two long parallel wires X and Y, separated by a distance of 6 cm, carry currents of 5A and 4A, respectively, in opposite directions as shown in the figure. Magnitude of the resultant magnetic field at point P at a distance of 4 cm from wire Y is $x \times 10^{-5}$ T. The value of x is _____. **[JEE-Mains Online-2025]**
Take permeability of free space as $\mu_0 = 4\pi \times 10^{-7}$ SI units.



92. A current of 5A exists in a square loop of side $\frac{1}{\sqrt{2}}$m. Then the magnitude of the magnetic field B at the centre of the square loop will be $p \times 10^{-6}$ T. where, value of p is _____. **[JEE-Mains Online-2025]**
[Take $\mu_0 = 4\pi \times 10^{-7}$ T mA^{-1}].

93. A conducting solid sphere of radius R and mass M carries a charge Q. The sphere is rotating about an axis passing through its center with a uniform angular speed ω. The ratio of the magnitudes of the magnetic dipole moment to the angular momentum about the same axis is given as $\alpha\frac{Q}{2M}$. The value of α is _____.

[JEE-Advanced-2025]

ANSWER KEY

EXERCISE-I

ONLY ONE OPTION IS CORRECT

1.	C	**2.**	A	**3.**	C	**4.**	B	**5.**	B	**6.**	B	**7.**	C
8.	A	**9.**	A	**10.**	A	**11.**	D	**12.**	B	**13.**	B	**14.**	A
15.	A	**16.**	A	**17.**	C	**18.**	B	**19.**	B	**20.**	A	**21.**	D
22.	A	**23.**	A	**24.**	C	**25.**	A	**26.**	B	**27.**	C	**28.**	B
29.	B	**30.**	C	**31.**	C	**32.**	C	**33.**	D	**34.**	A	**35.**	A
36.	A	**37.**	B	**38.**	B	**39.**	C	**40.**	B	**41.**	A	**42.**	B
43.	D	**44.**	A	**45.**	A	**46.**	B	**47.**	B	**48.**	A	**49.**	B
50.	B	**51.**	D	**52.**	D	**53.**	D	**54.**	C	**55.**	B	**56.**	D
57.	D												

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

58.	ABC	**59.**	ABC	**60.**	ABCD	**61.**	C	**62.**	BD	**63.**	AD	**64.**	BC
65.	CD	**66.**	D	**67.**	AB	**68.**	A	**69.**	ABD				

70. A – P,Q; B – P,R,S; C – P,S; D – P,Q,R,S

71. A – P,Q,R; B – P, Q, R, S ; C – R ; D – P, Q, R, S

EXERCISE-II

1. $\frac{\sqrt{2}\,\mu_0 i}{8\pi l}$

2. $\frac{\sqrt{5}}{2\sqrt{2}}\,4\pi \times 10^{-5}\,T$

3. $\frac{(2\sqrt{2}-1)\,\mu_0 i}{\pi a}$

4. 10.00

5. $\frac{\mu_0 I}{4R}\left(\frac{3}{4}\hat{k}+\frac{1}{\pi}\hat{j}\right)$

6. $B = \frac{\mu_0 i}{4\pi R}\sqrt{2\,(2\pi^2 - 2\pi + 1)}$

7. (i) 1.3×10^{-4} T, (ii) zero

8. μ_0 weber.m^{-1}

9. $\frac{\mu_0 iqv}{2\pi a}$

10. $10\hat{k}$

11. $\frac{2mv_0}{qB}$

12. zero

13. $F = \alpha a^2 i\,\hat{j}$

14. (a) $\frac{3mv^2}{4qa}$, (b) $\frac{3mv^3}{4a}$, (c) zero

15. In the plane of the drawing from right to left

16. $B_1 = \frac{\mu_0 b r_1^2}{3}$, $B_2 = \frac{\mu_0 b R^3}{3r_2}$

17. $T_0 = 2\pi\sqrt{\frac{m}{6IB}} = 0.57s$

18. $\frac{mEI}{Be}$

19. $\frac{\mu_0 I I' C}{2\pi}\left[\frac{1}{a}-\frac{1}{b}\right]$ to the left

20. $\sqrt{2}$ I R B

21. (i) $\frac{\mu_0}{4\pi}\left(\frac{4I}{a}\right)$ along Y-axis, (ii) $\frac{\mu_0}{4\pi}\left(\frac{I^2}{2a}\right)\sqrt{10}$, $\tan^{-1}\left(\frac{1}{3}\right)+\pi$ with positive axis

22. $\frac{mg}{2Il}$ **23.** (a) $\frac{QV}{m}\frac{\mu_0 I}{6a}\left(\frac{3\sqrt{3}}{\pi}-1\right)$, (b) $\vec{\tau}=BIa^2\left(\frac{\pi}{3}-\frac{\sqrt{3}}{4}\right)\hat{j}$

24. $\sqrt{15}$ C **25.** **(a)** $I=\frac{mg}{\pi r\left(B_x^2+B_y^2\right)^{1/2}}$ **(b)** $I=\frac{mg}{\pi r B_x}$

26. $\frac{\mu_0 iJ_0}{\pi}\tan^{-1}\left(\frac{d}{2h}\right)(-\hat{k})$ **27.** 0.62 N < F < 0.88 N **28.** B_{min} = 4.7 × 10^{-3} T

29. i_1 = 0.1110 A, i_2 = 0.096 A **30.** (a) $\vec{\tau}=\frac{BI_0L^2}{\sqrt{2}}\left(\hat{i}-\hat{j}\right)$ (b) $\theta=\frac{3}{4}\frac{BI_0}{M}\Delta t^2$

EXERCISE-III

1. D

2. (a) current in loop PQRS is clockwise from P to QRS., (b) $\vec{F}$ = BI_0b $(3\hat{k}-4\hat{i})$, (c) I = $\frac{mg}{6bB_0}$

3. $\omega=\frac{dT_0}{QR^2B}$ **4.** (a) k = NAB, (b) C = $\frac{2i_0NAB}{\pi}$, (c) Q × $\sqrt{\frac{NAB\pi}{2li_0}}$

5.	C	**6.**	B	**7.**	3.00	**8.**	ABC	**9.**	AD
10.	AC	**11.**	B	**12.**	ABD	**13.**	B	**14.**	ACD
15.	D	**16.**	D	**17.**	B	**18.**	B	**19.**	A
20.	C	**21.**	A	**22.**	A	**23.**	D	**24.**	C
25.	C	**26.**	C	**27.**	A	**28.**	B	**29.**	A
30.	D	**31.**	AC	**32.**	AB	**33.**	B	**34.**	A
35.	B	**36.**	B	**37.**	B	**38.**	C	**39.**	A
40.	C	**41.**	C	**42.**	A	**43.**	C	**44.**	C
45.	A	**46.**	B	**47.**	A	**48.**	628	**49.**	150
50.	40.00	**51.**	05.00	**52.**	02.00	**53.**	C	**54.**	B
55.	C	**56.**	A	**57.**	B	**58.**	B	**59.**	500
60.	20.00	**61.**	160	**62.**	5.00	**63.**	160	**64.**	4.00
65.	50	**66.**	A	**67.**	A,B	**68.**	A	**69.**	B
70.	B	**71.**	B	**72.**	B	**73.**	D	**74.**	A
75.	C	**76.**	B	**77.**	D	**78.**	A	**79.**	A
80.	C	**81.**	D	**82.**	A	**83.**	D	**84.**	D
85.	D	**86.**	C	**87.**	48.00	**88.**	10.00	**89.**	1.00
90.	2.00	**91.**	1.00	**92.**	8.00	**93.**	1.67		